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A MIXED FINITE ELEMENT FORMULATION OF TRIPHASIC MECHANO-ELECTROCHEMICAL THEORY FOR CHARGED, HYDRATED BIOLOGICAL SOFT TISSUES

D. N. SUN[1], W. Y. GU[2], X. E. GUO[1], W. M. LAI[1] AND V. C. MOW[1,*]

[1] *Orthopaedic Research Laboratory, Department of Mechanical Engineering and Orthopaedic Surgery, Columbia University, New York, NY 10032, U.S.A.*

[2] *Department of Biomedical Engineering, University of Miami, U.S.A.*

SUMMARY

An equivalent new expression of the triphasic mechano-electrochemical theory [9] is presented and a mixed finite element formulation is developed using the standard Galerkin weighted residual method. Solid displacement $\mathbf{u}^s$, *modified* electrochemical/chemical potentials ε^w, ε^+ and ε^- (with dimensions of concentration) for water, cation and anion are chosen as the four primary degrees of freedom (DOFs) and are independently interpolated. The modified Newton–Raphson iterative procedure is employed to handle the non-linear terms. The resulting first-order Ordinary Differential Equations (ODEs) with respect to time are solved using the implicit Euler backward scheme which is unconditionally stable. One-dimensional (1-D) linear isoparametric element is developed. The final algebraic equations form a non-symmetric but sparse matrix system. With the current choice of primary DOFs, the formulation has the advantage of small amount of storage, and the jump conditions between elements and across the interface boundary are satisfied automatically. The finite element formulation has been used to investigate a 1-D triphasic stress relaxation problem in the confined compression configuration and a 1-D triphasic free swelling problem. The formulation accuracy and convergence for 1-D cases are examined with independent finite difference methods. The FEM results are in excellent agreement with those obtained from the other methods. Copyright © 1999 John Wiley & Sons, Ltd.

KEY WORDS: soft tissue; triphasic mechano-electrochemical theory; finite element method

INTRODUCTION

Soft connective tissues are important constituents of the human body and their functions play critical roles in various normal and pathological conditions [1]. Although the morphology and the location of hydrated soft tissues such as articular cartilage, intervertebral disk, growth plate, vascular vessel, and skin are diverse, they share similarities in compositions. All such biological

* Correspondence to: V. C. Mow, Orthopaedic Research Laboratory, Columbia University, 630 West 168th Street BB1412, New York, NY 10032, U.S.A. E-mail:vcm1@columbia.edu

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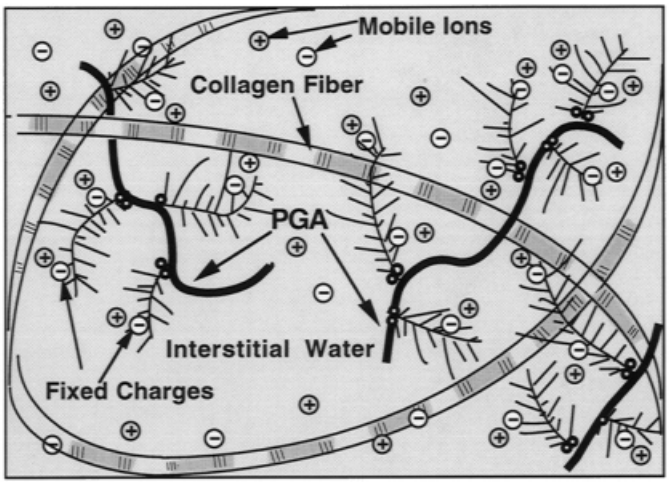

Figure 1. A schematic diagram of the structure of the charged hydrated articular cartilage

materials are multiphasic. In general, they are basically formed from a fibrous network of structure proteins (such as collagen and/or elastin with varying amounts of proteoglycons (PG), fibronectin and other glycoproteins), abundant interstitial water, soluble electrolytes such as NaCl and $CaCl_2$, etc., and cells residing in the interstitial space [2, 3]. The extracellular matrix (ECM) of connective tissues is porous and permeable [4]. Often, there are negatively or positively charged groups on the ECM, or the solid 'skeleton' of the soft tissues. These charges on the solid ECM are called fixed charges because their mobility is much less than that of freely mobile ions dissolved in the interstitial water. Articular cartilage, for example, is such a hydrated, charged soft tissue. It consists mostly of dense network of type II collagen fibers and PG aggrecans; this fiber-gel forms the porous-permeable matrix. Within the interstitial fluid, at physiological pH and ionic strength, PG aggrecans, with abundant sulphate groups on their glycosaminoglycan (GAG) chains, become negatively charged and attract mobile cation ions (e.g. Na^+) in the interstitial fluid to maintain the electroneutrality of the tissue as a whole (Figure 1). The mechano-electrochemical properties of such tissues depend on the relative amount of the tissue components, ultrastructure organization of the solid matrix and the mechano-electrical interaction of the components within the extracellular matrix. Many mechano-electrochemical coupling phenomena such as ion-induced swelling, streaming potential and current, anomalous osmosis and electro-osmosis observed in such charged biological tissues are attributed to the electrostatic and frictional interactions between the fixed charges and mobile ions.

A number of theoretical models have been developed to investigate the flow-dependent viscoelastic behavior (e.g. [4–7]) and mechano-electrochemical behavior (e.g. [8–11]) of charged hydrated soft tissues, using (1) the continuum mixture theory approach ([12, 13, 59], also see recent review paper by de Boer [14]); (2) the consolidation theory approach [15, 16]; or (3) the combination of biphasic mixture theory [4] or the consolidation theory [15, 16] with a phenomenological transport theory [17]. Due to the complexity of the above theories for the multiphasic materials, it is impossible to obtain an analytical solution for most of the practical boundary-value problems associated with specific experiments. Thus, the finite element method, because of

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its ability to handle complicated geometry and boundary conditions, becomes the best choice of all the numerical tools.

The finite element models for the biphasic theory [4, 18–20] were developed by Spilker and co-workers. The penalty formulation [21, 22], mixed penalty formulation [23], hybrid formulation [24] and mixed u–p formulation [25, 26] were proposed. With the progress of biphasic finite element method, a variety of problems such as friction effect on the unconfined compression [27], indentation analysis under finite deformation [28] and non-isotropic analysis [29] have been solved. Some challenging topics such as automated-mesh-adaptivity [30] and joint contact problems [31, 32] are being addressed. Recently, some investigators began to describe the contributions of ions using finite element method [11, 33–35].

It should be pointed out here that, according to the interface jump conditions [36], the fluid velocity can be discontinuous across the interface between two biphasic layers with different porosities or between a biphasic layer and a single phase layer. In other words, with different porosities on both sides, the fluid velocity has no unique value right at the interface. Thus, if the fluid velocity is chosen as one of the primary degrees of freedom, the finite element formulation will be limited only to the problems in which the intrinsic porosity of the material is continuous.

A triphasic mechano-electrochemical theory has been developed by Lai *et al.* [9], which encompasses both the biphasic view and physico-chemical view of the soft tissues. It is also consistent with and subsumes many previously derived mixture and phenomenological transport theories pertaining to specific aspects of the charged-hydrated tissues (e.g. [37–43]). It has been used to describe mechano-electrochemical behavior of the soft tissues such as cartilage and intervertebral disk [9, 44–48]. The objectives of this paper are: (1) to derive a concise form of the triphasic mechano-electrochemical theory [9, 48] suitable for finite element formulation; (2) to develop a finite element formulation for the triphasic theory; (3) to test the accuracy and the convergence of the formulation by comparison of finite element solutions with independent finite difference solutions. As a first attempt to develop the finite element formulation for the triphasic theory and without losing generality, only the monovalent (1:1) electrolyte and infinitesimal deformation are considered here.

THE TRIPHASIC MECHANO-ELECTROCHEMICAL THEORY

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The triphasic mechano-electrochemical theory [9] considers a charged hydrated soft tissue as a mixture consisting of: (1) an intrinsically incompressible, porous-permeable-charged solid phase (superscript s); (2) an intrinsically incompressible interstitial fluid phase (superscript w); and (3) ion phase with two monovalent ion species, i.e., anion (−) and cation (+). Recently, this theory has been extended to include multiple species of ions [48]. We will only consider 1:1 electrolyte case here while the method can be extended to include the multiple ion cases without difficulty. This theory is strictly based on the laws of continuum mechanics and the second law of thermodynamics.

Saturation condition

Let ϕ^α denote the volume fraction of each component (i.e., $\phi^\alpha = \mathrm{d}V^\alpha/\mathrm{d}V$, $\alpha = \mathrm{s}, \mathrm{w}, +, -$), we have the saturation condition

$$\phi^{\mathrm{s}} + \phi^{\mathrm{w}} + \phi^{+} + \phi^{-} = 1 \tag{1}$$

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where $\phi^{\mathrm{s}} = \mathrm{d}V^{\mathrm{s}}/\mathrm{d}V$ is the solidity, $\phi^{\mathrm{w}} = \mathrm{d}V^{\mathrm{w}}/\mathrm{d}V$ is the porosity of the tissue, and ϕ^{+} and ϕ^{-} are ion volume fractions. Compared with ϕ^{s} and ϕ^{w}, ϕ^{+} and ϕ^{-} are usually very small and can be neglected practically.

Continuity equations and electroneutrality condition

When no chemical reaction takes place, conservation of mass for each component requires

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$$\frac{\partial \rho^{\alpha}}{\partial t} + \mathrm{div}(\rho^{\alpha}\mathbf{v}^{\alpha}) = 0, \quad (\alpha = \mathrm{s}, \mathrm{w}, +, -) \tag{2}$$

where $\mathbf{v}^{\alpha}$ is the velocity of α component, ρ^{α} is its mass density, related to true solid density $\rho_{\mathrm{T}}^{\mathrm{s}}$, true water density $\rho_{\mathrm{T}}^{\mathrm{w}}$, cation molar concentration c^{+}, anion molar concentration c^{-}, solidity ϕ^{s} and porosity ϕ^{w} by

$$\rho^{s} = \rho_{\mathrm{T}}^{\mathrm{s}}\phi^{\mathrm{s}} \tag{3}$$

$$\rho^{\mathrm{w}} = \rho_{\mathrm{T}}^{\mathrm{w}}\phi^{\mathrm{w}} \tag{4}$$

$$\rho^{+} = \phi^{\mathrm{w}}c^{+}M_{+} = \rho_{\mathrm{T}}^{+}\phi^{+} \tag{5}$$

$$\rho^{-} = \phi^{\mathrm{w}}c^{-}M_{-} = \rho_{\mathrm{T}}^{-}\phi^{-} \tag{6}$$

where M_{+} and M_{-} are molecular weight of cation and anion, respectively.

The porosity is the volume fraction of the extrafibrillar water which is available to the charges on the PGs [49]. The intrafibrillar water within the collagen fibers, therefore, may be regarded as bounded water and thus part of the solid phase [50]. From (1)–(6), assuming the true densities are constant, the continuity equation for the mixture can be written as

$$\mathrm{div}(\phi^{\mathrm{s}}\mathbf{v}^{\mathrm{s}} + \phi^{\mathrm{w}}\mathbf{v}^{\mathrm{w}} + \phi^{+}\mathbf{v}^{+} + \phi^{-}\mathbf{v}^{-}) = 0 \tag{7}$$

or by neglecting ϕ^{+} and ϕ^{-}(when compared with ϕ^{s} and ϕ^{w}), we have

$$\mathrm{div}(\phi^{\mathrm{s}}\mathbf{v}^{\mathrm{s}} + \phi^{\mathrm{w}}\mathbf{v}^{\mathrm{w}}) = 0 \tag{8}$$

Because of intrinsical incompressibility, for infinitesimal strain, ϕ^{s} can be expressed as

$$\phi^{\mathrm{s}} = \phi_{0}^{\mathrm{s}}/(1 + e) \tag{9}$$

where ϕ_{0}^{s} is the tissue solidity at a physicochemical reference configuration. Note that it is convenient theoretically to choose the hypertonic state (corresponding to infinite salt concentration) as a reference configuration, while practically it is convenient to choose the free swelling (or load-free) state as a reference configuration. The dilatation e is related to the infinitesimal strain tensor of the solid matrix $\mathbf{E}$ by $e = \mathrm{tr}(\mathbf{E})$.

The electroneutrality condition can be written as

$$c^{+} = c^{-} + c^{\mathrm{F}} \tag{10}$$

where c^{F} is the Fixed Charge Density (FCD) measured as equivalent moles per unit volume of water. In equation (10) we have assumed the solid matrix to be negatively charged. Since in the triphasic theory, the fixed charges on the solid phase are assumed to remain unchanged, c^{F} must

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be conserved at all time during deformation of the tissue. Thus

$$\partial(\phi^{\mathrm{w}}c^{\mathrm{F}})/\partial t + \operatorname{div}(\phi^{\mathrm{w}}c^{\mathrm{F}}\mathbf{v}^{\mathrm{s}}) = 0 \tag{11}$$

or, it can be expressed with the tissue deformation as

$$c^{\mathrm{F}} = c_0^{\mathrm{F}}/(1 + e/\phi_0^{\mathrm{w}}) \tag{12}$$

where c_0^{F} and ϕ_0^{w} are the FCD and water content (porosity) of the tissue at the reference state. The conservation of ions and fixed charges, and the electroneutrality condition (10) yield

$$\operatorname{div}(\mathbf{I}_{\mathrm{e}}) = 0 \tag{13}$$

where $\mathbf{I}_{\mathrm{e}}$ is the electrical current density given by

$$\mathbf{I}_{\mathrm{e}} = F_{\mathrm{c}}\phi^{\mathrm{w}}[c^{+}(\mathbf{v}^{+} - \mathbf{v}^{\mathrm{s}}) - c^{-}(\mathbf{v}^{-} - \mathbf{v}^{\mathrm{s}})] \tag{14}$$

where F_{c} is the Faraday constant.

Momentum equations

In the triphasic theory, by neglecting gravitational and magnetic effects, the gradient of the chemical/electrochemical potentials are the motive (driving) forces, balanced by the frictional forces between one phase and another. Thus, the momentum equations are

Tissue
$$\nabla\cdot\boldsymbol{\sigma} = 0 \tag{15}$$

Water
$$-\rho^{\mathrm{w}}\nabla\mu^{\mathrm{w}} + f_{\mathrm{ws}}(\mathbf{v}^{\mathrm{s}} - \mathbf{v}^{\mathrm{w}}) + f_{\mathrm{w}+}(\mathbf{v}^{+} - \mathbf{v}^{\mathrm{w}}) + f_{\mathrm{w}-}(\mathbf{v}^{-} - \mathbf{v}^{\mathrm{w}}) = 0 \tag{16}$$

Cation
$$-\rho^{+}\nabla\tilde{\mu}^{+} + f_{+\mathrm{s}}(\mathbf{v}^{\mathrm{s}} - \mathbf{v}^{+}) + f_{+w}(\mathbf{v}^{\mathrm{w}} - \mathbf{v}^{+}) + f_{+-}(\mathbf{v}^{-} - \mathbf{v}^{+}) = 0 \tag{17}$$

Anion
$$-\rho^{-}\nabla\tilde{\mu}^{-} + f_{-\mathrm{s}}(\mathbf{v}^{\mathrm{s}} - \mathbf{v}^{-}) + f_{-\mathrm{w}}(\mathbf{v}^{\mathrm{w}} - \mathbf{v}^{-}) + f_{-+}(\mathbf{v}^{+} - \mathbf{v}^{-}) = 0 \tag{18}$$

where $\boldsymbol{\sigma}$ is the total stress of the mixture (tissue), $\tilde{\mu}^{+}$and $\tilde{\mu}^{-}$ are electrochemical potentials for cation and anion, respectively, and μ^{w} is the chemical potential for water. Here, $f_{\alpha\beta}$'s are frictional coefficients per unit tissue volume between the inter-diffusing α and β components. In the triphasic theory, the frictional coefficients per unit tissue volume are assumed to be symmetric, i.e. $f_{\alpha\beta} = f_{\beta\alpha}$, which is consistent with Onsager reciprocity relationship [37, 38].

Constitutive equations

For an isotropic hydrated charged mixture with infinitesimal deformation, the constitutive equations of the triphasic theory are

$$\boldsymbol{\sigma} = -p\mathbf{I} - T_{\mathrm{c}}\mathbf{I} + \sigma^{\mathrm{e}} = -p\mathbf{I} - T_{\mathrm{c}}\mathbf{I} + \lambda_{\mathrm{s}}e\mathbf{I} + 2\mu_{\mathrm{s}}\mathbf{E} \tag{19}$$

$$\mu^{\mathrm{w}} = \mu_0^{\mathrm{w}} + [p - RT\phi(c^{+} + c^{-}) + B_{\mathrm{w}}e]/\rho_{\mathrm{T}}^{\mathrm{w}} \tag{20}$$

$$\tilde{\mu}^{+} = \mu_0^{+} + (RT/M_{+})\ln(\gamma_{+}c^{+}) + F_{\mathrm{c}}\psi/M_{+} \tag{21}$$

$$\tilde{\mu}^{-} = \mu_0^{-} + (RT/M_{-})\ln(\gamma_{-}c^{-}) - F_{\mathrm{c}}\psi/M_{-} \tag{22}$$

$$T_{\mathrm{c}} = a_0c^{\mathrm{F}}\exp\{-\kappa(\gamma_{\pm}/\gamma_{\pm}^{*})[c^{+}c^{-}]^{1/2}\} \tag{23}$$

The definitions of the physical quantities and symbols used above are listed in Table I.

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Table I. The parameter definitions of the triphasic theory

Quantities and Symbols	Definitions
p	Fluid pressure
T_c	Chemical expansion stress
σ^e	Elastic part of the solid matrix stress
λ_s, μ_s	Lame coefficients of the solid matrix
μ_0^α $(\alpha = \mathrm{w}, +, -)$	Reference chemical potentials
γ_+, γ_-	Activity coefficients of cation and anion
$\gamma_\pm = (\gamma_+\gamma_-)^{1/2}$	Mean activity coefficient of the salt
ϕ	Osmotic coefficient
ψ	Electrical potential
c_0^F	FCD at the reference configuration
ϕ_0^w	Tissue porosity at the reference configuration
a_0, κ, B_w	Material constants
R	Universal gas constant
T	Absolute temperature
F_c	Faraday constant
$*$	Refers to the quantities in external solutions

AN EQUIVALENT FORMULATION FOR THE TRIPHASIC THEORY

In the following, an equivalent formulation for the triphasic theory will be derived. This new formulation permits easy implementation for finite element realization and has the advantage that the jump conditions between elements and across the interface boundary are satisfied automatically even for inhomogeneous tissues with discontinuous material properties.

First, using a simple transformation, the modified electrochemical/chemical potential functions are defined by:

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E_w

E_p

E_n

$$\varepsilon^{\mathrm{w}} = \frac{\rho_{\mathrm{T}}^{\mathrm{w}}(\mu^{\mathrm{w}} - \mu_0^{\mathrm{w}})}{RT} = \frac{p}{RT} - \phi(c^+ + c^-) + \frac{B_{\mathrm{w}}}{\mathbf{RT}} e \tag{24}$$

$$\varepsilon^+ = \exp\left[\frac{M^+(\tilde{\mu}^+ - \tilde{\mu}_0^+)}{RT}\right] = \gamma_+ c^+ \exp\left(\frac{F_c\psi}{RT}\right) \tag{25}$$

$$\varepsilon^- = \exp\left[\frac{M^-(\tilde{\mu}^- - \tilde{\mu}_0^-)}{RT}\right] = \gamma_- c^- \exp\left(-\frac{F_c\psi}{RT}\right) \tag{26}$$

Note that ε^{w}, ε^+ and ε^- all have a dimension of ion concentration. The above definition avoids a potential mathematical complexity of logarithm in numerical implementation [35].

The water and ion fluxes relative to the solid phase are defined by

$$\mathbf{J}^{\mathrm{w}} = \phi^{\mathrm{w}}(\mathbf{v}^{\mathrm{w}} - \mathbf{v}^{\mathrm{s}}) \tag{27}$$

$$\mathbf{J}^+ = \phi^{\mathrm{w}} c^+ (\mathbf{v}^+ - \mathbf{v}^{\mathrm{s}}) \tag{28}$$

$$\mathbf{J}^- = \phi^{\mathrm{w}} c^- (\mathbf{v}^- - \mathbf{v}^{\mathrm{s}}) \tag{29}$$

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Note that with the above definitions, the solid displacement $\mathbf{u}^s$ (for infinitesimal deformation, it is related to the solid matrix velocity $\mathbf{v}^s$ as $\mathbf{v}^s = \partial\mathbf{u}^s/\partial t$) and modified chemical potentials ε^w, ε^+ and ε^- compose the four primary Degrees of freedom (DOFs) of the whole system, all the other unknowns can be derived from them (see Appendix I). Also with the triphasic momentum equations (15)–(17), all the fluxes can be expressed with the combination of the gradients of the chemical potentials as follows:

$$\mathbf{J}^w = -\frac{RT\phi^w}{\alpha}\left(\nabla\varepsilon^w + \frac{c^+}{\varepsilon^+}\nabla\varepsilon^+ + \frac{c^-}{\varepsilon^-}\nabla\varepsilon^-\right) \tag{30}$$

$$\mathbf{J}^+ = -\frac{RT\phi^w c^+}{\alpha}\nabla\varepsilon^w - \left[\frac{\phi^w c^+ D^+}{\varepsilon^+} + \frac{RT\phi^w (c^+)^2}{\alpha\varepsilon^+}\right]\nabla\varepsilon^+ - \frac{RT\phi^w c^+ c^-}{\alpha\varepsilon^-}\nabla\varepsilon^- \tag{31}$$

$$\mathbf{J}^- = -\frac{RT\phi^w c^-}{\alpha}\nabla\varepsilon^w - \frac{RT\phi^w c^+ c^-}{\alpha\varepsilon^+}\nabla\varepsilon^+ - \left[\frac{\phi^w c^- D^-}{\varepsilon^-} + \frac{RT\phi^w (c^-)^2}{\alpha\varepsilon^-}\right]\nabla\varepsilon^- \tag{32}$$

Note that in the above derivation, the following relations [17, 45] were used

$$f_{sw} = \alpha\phi^w \text{ (or, } \alpha = f_{sw}/\phi^w) \tag{33}$$

$$f_{+w} = RT\phi^w c^+/D^+ \tag{34}$$

$$f_{-w} = RT\phi^w c^-/D^- \tag{35}$$

where D^+ and D^- are the diffusivities of the cation and anion, respectively. α, D^+ and D^- were assumed to be constants. In the triphasic theory, the hydraulic permeability k is related with α by

$$k = \phi^w/\alpha = (\phi^w)^2/f_{ws} \tag{36}$$

and f_{+s}, f_{-s}, f_{+-}, f_{-+} are neglected [10, 45].

Governing equations

With $\mathbf{u}^s$, ε^w, ε^+ and ε^- as the primary DOFs, and the definitions of the fluxes (27)–(29), we have the following equations:

momentum equation for the mixture

$$\nabla\cdot\boldsymbol{\sigma} = 0 \tag{37}$$

continuity equation of the mixture

$$\nabla\cdot\mathbf{v}^s + \nabla\cdot\mathbf{J}^w = 0 \tag{38}$$

continuity equation for ions

$$\partial(\phi^w c^+)/\partial t + \nabla\cdot\mathbf{J}^+ + \nabla\cdot(\phi^w c^+\mathbf{v}^s) = 0 \tag{39}$$

$$\partial(\phi^w c^-)/\partial t + \nabla\cdot\mathbf{J}^- + \nabla\cdot(\phi^w c^-\mathbf{v}^s) = 0 \tag{40}$$

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Using the electroneutrality condition (10), one can get the following equivalent equations to substitute in equation (39) or (40):

electrical current condition

$$\nabla \cdot \mathbf{J}^{+} - \nabla \cdot \mathbf{J}^{-} = 0 \tag{41}$$

ion diffusion-convection equation

$$\partial(\phi^{\mathrm{w}} c^{k})/\partial t + \nabla \cdot \mathbf{J}^{+} + \nabla \cdot \mathbf{J}^{-} + \nabla \cdot (\phi^{\mathrm{w}} c^{k} \mathbf{v}^{\mathrm{s}}) = 0 \tag{42}$$

where

$$c^{k} = c^{+} + c^{-} \tag{43}$$

Substituting equations (19), (30)–(32) and (15) into equations (37), (38), (41) and (42), we get the following governing equations in terms of strain tensor (or solid displacement) and modified electrochemical/chemical potentials:

$$\nabla \cdot (\lambda \mathrm{e}\mathbf{I} + 2\mu_{\mathrm{s}} \mathbf{E}) - \nabla (RT\varepsilon^{\mathrm{w}} + RT\phi c^{k} - B_{\mathrm{w}} e) = 0 \tag{44}$$

$$\nabla \cdot \mathbf{v}^{\mathrm{s}} - \nabla \cdot \frac{RT}{\alpha} \left[\phi^{\mathrm{w}} \nabla \varepsilon^{\mathrm{w}} + \frac{\phi^{\mathrm{w}} c^{+}}{\varepsilon^{+}} \nabla \varepsilon^{+} + \frac{\phi^{\mathrm{w}} c^{-}}{\varepsilon^{-}} \nabla \varepsilon^{-} \right] = 0 \tag{45}$$

$$\begin{aligned} \nabla \cdot \Bigg\{ & -\frac{RT}{\alpha} \phi^{\mathrm{w}} c^{\mathrm{F}} \nabla \varepsilon^{\mathrm{w}} - \left[\frac{\phi^{\mathrm{w}} c^{+} D^{+}}{\varepsilon^{+}} + \frac{RT}{\alpha} \frac{\phi^{\mathrm{w}} (c^{+})^{2}}{\varepsilon^{+}} - \frac{RT}{\alpha} \frac{\phi^{w} c^{+} c^{-}}{\varepsilon^{+}} \right] \nabla \varepsilon^{+} \\ & + \left[\frac{\phi^{\mathrm{w}} c^{-} D^{-}}{\varepsilon^{-}} + \frac{RT}{\alpha} \frac{\phi^{\mathrm{w}} (c^{-})^{2}}{\varepsilon^{-}} - \frac{RT}{\alpha} \frac{\phi^{\mathrm{w}} c^{+} \mathrm{c}^{-}}{\varepsilon^{-}} \right] \nabla \varepsilon^{-} \Bigg\} = 0 \end{aligned} \tag{46}$$

$$\begin{aligned} \partial(\phi^{\mathrm{w}} c^{k})/\partial t - \nabla \cdot \Bigg\{ & \frac{RT}{\alpha} \phi^{\mathrm{w}} c^{k} \nabla \varepsilon^{\mathrm{w}} + \left[\frac{\phi^{\mathrm{w}} c^{+} D^{+}}{\varepsilon^{+}} + \frac{RT}{\alpha} \frac{\phi^{\mathrm{w}} (c^{+})^{2}}{\varepsilon^{+}} + \frac{RT}{\alpha} \frac{\phi^{\mathrm{w}} c^{+} c^{-}}{\varepsilon^{+}} \right] \nabla \varepsilon^{+} \\ & + \left[\frac{\phi^{\mathrm{w}} c^{-} D^{-}}{\varepsilon^{-}} + \frac{RT \phi^{\mathrm{w}} (c^{-})^{2}}{\alpha \quad \varepsilon^{-}} + \frac{RT}{\alpha} \frac{\phi^{\mathrm{w}} c^{+} c^{-}}{\varepsilon^{-}} \right] \nabla \varepsilon^{-} \Bigg\} + \nabla \cdot (\phi^{\mathrm{w}} c^{k} \mathbf{v}^{\mathrm{s}}) = 0 \end{aligned} \tag{47}$$

Notice that for infinitesimal strain the chemical expansion stress T_{c} has been neglected in equations (44)–(47) [47, 51, 52].

Boundary conditions

Based on the jump conditions across the boundaries or material interface (see Appendix II), the following boundary conditions can be prescribed as

$$\mathbf{u}^{\mathrm{s}} = \mathbf{u}^{\mathrm{s}*} \qquad \text{on } \Gamma_{u} \tag{48}$$

$$\varepsilon^{\mathrm{w}} = \varepsilon^{\mathrm{w}*} \qquad \text{on } \Gamma_{\varepsilon^{\mathrm{w}}} \tag{49}$$

$$\varepsilon^{+} = \varepsilon^{+*} \qquad \text{on } \Gamma_{\varepsilon^{+}} \tag{50}$$

$$\varepsilon^{-} = \varepsilon^{-*} \qquad \text{on } \Gamma_{\varepsilon^{-}} \tag{51}$$

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or

$$\boldsymbol{\sigma}\cdot\mathbf{n} = \boldsymbol{\sigma}^{*}\cdot\mathbf{n} \qquad \text{on } \Gamma_t \tag{52}$$

$$\mathbf{J}^{\mathrm{w}}\cdot\mathbf{n} = \mathbf{J}^{\mathrm{w}*}\cdot\mathbf{n} \qquad \text{on } \Gamma_{J^{\mathrm{w}}} \tag{53}$$

$$\mathbf{J}^{+}\cdot\mathbf{n} = \mathbf{J}^{+*}\cdot\mathbf{n} \qquad \text{on } \Gamma_{J^{+}} \tag{54}$$

$$\mathbf{J}^{-}\cdot\mathbf{n} = \mathbf{J}^{-*}\cdot\mathbf{n} \qquad \text{on } \Gamma_{J^{-}} \tag{55}$$

where $\mathbf{n}$ is the outward unit normal to the boundary, and * indicates the prescribed values.

In the following finite element formulation, boundary conditions (48)–(51) belong to essential boundary conditions while equations (52)–(55) involve the combination of the gradients of the primary DOFs and represent natural boundary conditions.

THE FINITE ELEMENT FORMULATION

Non-dimensionalization

Let h, c_0 and D_0 be the basic scales of length, concentration and diffusivity, respectively, then the non-dimensional quantities (denoted by subscript r) are defined by

$$\begin{gathered} x_r = \frac{x}{h}, \quad \mathbf{u}_{\mathrm{r}}^{\mathrm{s}} = \frac{\mathbf{u}^{\mathrm{s}}}{h}, \quad \mathbf{v}_{\mathrm{r}}^{\mathrm{s}} = \frac{\mathbf{v}^{\mathrm{s}}}{v_0}, \quad t_r = \frac{t}{\tau} \\ D_{\mathrm{r}}^{\alpha} = \frac{D^{\alpha}}{D_0}, \quad c_{\mathrm{r}}^{\alpha} = \frac{c^{\alpha}}{c_0}, \quad \sigma_{\mathrm{r}} = \frac{\sigma}{\sigma_0}, \quad \varepsilon_{\mathrm{r}}^{\alpha} = \frac{\varepsilon^{\alpha}}{c_0} \\ \mathbf{J}_{\mathrm{r}}^{\mathrm{w}} = \frac{\mathbf{J}^{\mathrm{w}}}{v_0}, \quad \mathbf{J}_{\mathrm{r}}^{+} = \frac{\mathbf{J}^{+}}{c_0 v_0}, \quad \mathbf{J}_{\mathrm{r}}^{-} = \frac{\mathbf{J}^{-}}{c_0 v_0} \end{gathered} \tag{56}$$

where

$$v_0 = D_0/h, \quad \tau = h/v_0 = h^2/D_0, \quad \sigma_0 = RTc_0. \tag{57}$$

Correspondingly, the governing equations and boundary conditions can be non-dimensionalized as follows:

$$\nabla_{\mathrm{r}}\cdot\boldsymbol{\sigma}_{\mathrm{r}} = 0 \tag{58}$$

$$\nabla_{\mathrm{r}}\cdot\mathbf{v}_{\mathrm{r}}^{\mathrm{s}} - \nabla_{\mathrm{r}}\cdot k_D\left[\phi^{w}\nabla_{\mathrm{r}}\varepsilon_{\mathrm{r}}^{\mathrm{w}} + \frac{\phi^{\mathrm{w}}c_{\mathrm{r}}^{+}}{\varepsilon_{\mathrm{r}}^{+}}\nabla_{\mathrm{r}}\varepsilon_{\mathrm{r}}^{+} + \frac{\phi^{\mathrm{w}}c_{\mathrm{r}}^{-}}{\varepsilon_{\mathrm{r}}^{-}}\nabla_{\mathrm{r}}\varepsilon_{\mathrm{r}}^{-}\right] = 0 \tag{59}$$

$$\begin{aligned} \nabla_r\cdot\Bigg\{ &- k_D\phi^{\mathrm{w}}c_{\mathrm{r}}^{\mathrm{F}}\nabla_{\mathrm{r}}\varepsilon_{\mathrm{r}}^{\mathrm{w}} - \left[\frac{\phi^{\mathrm{w}}c_{\mathrm{r}}^{+}D_{\mathrm{r}}^{+}}{\varepsilon_{\mathrm{r}}^{+}} + k_{\mathrm{D}}\frac{\phi^{\mathrm{w}}(c_{\mathrm{r}}^{+})^2}{\varepsilon_{\mathrm{r}}^{+}} - k_{\mathrm{D}}\frac{\phi^{\mathrm{w}}c_{\mathrm{r}}^{+}c_{\mathrm{r}}^{-}}{\varepsilon_{\mathrm{r}}^{+}}\right]\nabla_{\mathrm{r}}\varepsilon_{\mathrm{r}}^{+} \\ &+ \left[\frac{\phi^{\mathrm{w}}c_{\mathrm{r}}^{-}D_{\mathrm{r}}^{-}}{\varepsilon_{\mathrm{r}}^{-}} + k_{\mathrm{D}}\frac{\phi^{\mathrm{w}}(c_{\mathrm{r}}^{-})^2}{\varepsilon_{\mathrm{r}}^{-}} - k_{\mathrm{D}}\frac{\phi^{\mathrm{w}}c_{\mathrm{r}}^{+}c_{\mathrm{r}}^{-}}{\varepsilon_{\mathrm{r}}^{-}}\right]\nabla_{\mathrm{r}}\varepsilon_{\mathrm{r}}^{-}\Bigg\} = 0 \end{aligned} \tag{60}$$

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$$\frac{\partial(\phi^{\mathrm{w}}c_{\mathrm{r}}^{\mathrm{k}})}{\partial t_{\mathrm{r}}} - \nabla_{\mathrm{r}}\cdot\left\{k_D\phi^{\mathrm{w}}c_{\mathrm{r}}^{k}\nabla_{\mathrm{r}}\varepsilon_{\mathrm{r}}^{\mathrm{w}} + \left[\frac{\phi^{\mathrm{w}}c_{\mathrm{r}}^{+}D_{\mathrm{r}}^{+}}{\varepsilon_{\mathrm{r}}^{+}} + k_{\mathrm{D}}\frac{\phi^{\mathrm{w}}(c_{\mathrm{r}}^{+})^2}{\varepsilon_{\mathrm{r}}^{+}} + k_{\mathrm{D}}\frac{\phi^{\mathrm{w}}c_{\mathrm{r}}^{+}c_{\mathrm{r}}^{-}}{\varepsilon_{\mathrm{r}}^{+}}\right]\nabla_{\mathrm{r}}\varepsilon_{\mathrm{r}}^{+}\right.$$
$$\left. + \left[\frac{\phi^{\mathrm{w}}c_{\mathrm{r}}^{-}D_{\mathrm{r}}^{-}}{\varepsilon_{\mathrm{r}}^{-}} + k_{\mathrm{D}}\frac{\phi^{\mathrm{w}}(c_{\mathrm{r}}^{-})^2}{\varepsilon_{\mathrm{r}}^{-}} + k_{\mathrm{D}}\frac{\phi^{\mathrm{w}}c_{\mathrm{r}}^{+}c_{\mathrm{r}}^{-}}{\varepsilon_{\mathrm{r}}^{-}}\right]\nabla_{\mathrm{r}}\varepsilon_{\mathrm{r}}^{-}\right\} + \nabla_{\mathrm{r}}\cdot(\phi^{\mathrm{w}}c_{\mathrm{r}}^{k}\mathbf{v}_{\mathrm{r}}^{\mathrm{s}}) = 0 \tag{61}$$

where the non-dimensional parameter (k_D) is $k_D = RTc_0/\alpha D_0$.

The finite element formulation

Using the standard Galerkin weighted residual method [53, 54] the finite element formulation is constructed. Let vector $\mathbf{w}$ and three scalar functions g, θ, and s be arbitrary admissible weighting functions for the four non-dimensional governing equations, i.e., mixture momentum equation, the mixture continuity equation, the current condition and the ion diffusion–convection equation, respectively. The weak formulation of triphasic theory is as follows:

$$\int_\Omega \mathbf{w}\cdot\nabla\cdot\boldsymbol{\sigma}\,\mathrm{d}\Omega = 0 \tag{62}$$

$$\int_\Omega g\left\{\nabla\cdot\mathbf{v}^{\mathrm{s}} - \nabla\cdot k_D\left[\phi^{\mathrm{w}}\nabla\varepsilon^{\mathrm{w}} + \frac{\phi^{\mathrm{w}}c^{+}}{\varepsilon^{+}}\nabla\varepsilon^{+} + \frac{\phi^{\mathrm{w}}c^{-}}{\varepsilon^{-}}\nabla\varepsilon^{-}\right]\right\}\mathrm{d}\Omega = 0 \tag{63}$$

$$\int_\Omega \theta\nabla\cdot\left\{-k_D\phi^{\mathrm{w}}c^{\mathrm{F}}\nabla\varepsilon^{\mathrm{w}} - \left[\frac{\phi^{\mathrm{w}}c^{+}D^{+}}{\varepsilon^{+}} + k_{\mathrm{D}}\frac{\phi^{\mathrm{w}}(c^{+})^2}{\varepsilon^{+}} - k_{\mathrm{D}}\frac{\phi^{\mathrm{w}}c^{+}c^{-}}{\varepsilon^{+}}\right]\nabla\varepsilon^{+}\right.$$
$$\left. + \left[\frac{\phi^{\mathrm{w}}c^{-}D^{-}}{\varepsilon^{-}} + k_{\mathrm{D}}\frac{\phi^{\mathrm{w}}(c^{-})^2}{\varepsilon^{-}} - k_{\mathrm{D}}\frac{\phi^{\mathrm{w}}c^{+}c^{-}}{\varepsilon^{-}}\right]\nabla\varepsilon^{-}\right\}\mathrm{d}\Omega = 0 \tag{64}$$

$$\int_\Omega s\left(\partial(\phi^{\mathrm{w}}c^{\mathrm{k}})/\partial t - \nabla\cdot\left\{k_D\phi^{\mathrm{w}}c^{\mathrm{k}}\nabla\varepsilon^{\mathrm{w}} + \left[\frac{\phi^{\mathrm{w}}c^{+}D^{+}}{\varepsilon^{+}} + k_D\frac{\phi^{\mathrm{w}}(c^{+})^2}{\varepsilon^{+}} + k_D\frac{\phi^{\mathrm{w}}c^{+}c^{-}}{\varepsilon^{+}}\right]\nabla\varepsilon^{+}\right.\right.$$
$$\left.\left. + \left[\frac{\phi^{\mathrm{w}}c^{-}D^{-}}{\varepsilon^{-}} + k_D\frac{\phi^{\mathrm{w}}(c^{-})^2}{\varepsilon^{-}} + k_D\frac{\phi^{\mathrm{w}}c^{+}c^{-}}{\varepsilon^{-}}\right]\nabla\varepsilon^{-}\right\} + \nabla\cdot(\phi^{\mathrm{w}}c^{k}\mathbf{v}^{\mathrm{s}})\right)\mathrm{d}\Omega = 0 \tag{65}$$

The subscript r has been dropped for simplicity in all the non-dimensional equations here and below. Applying the divergence theorem, we obtain

$$\int_\Omega \mathrm{tr}\,[(\nabla\mathbf{w})^{\mathrm{T}}\cdot\boldsymbol{\sigma}]\,\mathrm{d}\Omega = \int_{\Gamma_t}\mathbf{w}\cdot\mathbf{t}^*\,\mathrm{d}\Gamma \tag{66}$$

$$\int_\Omega g\nabla\cdot\mathbf{v}^{\mathrm{s}}\mathrm{d}\Omega + \int_\Omega k_D\phi^{\mathrm{w}}\left(\nabla\varepsilon^{\mathrm{w}} + \frac{c^{+}}{\varepsilon^{+}}\nabla\varepsilon^{+} + \frac{c^{-}}{\varepsilon^{-}}\nabla\varepsilon^{-}\right)\cdot\nabla g\,\mathrm{d}\Omega = -\int_{\Gamma_{J^w}} g\mathbf{J}^{\mathrm{w}*}\cdot\mathbf{n}\,\mathrm{d}\Gamma \tag{67}$$

$$\int_\Omega (k_D\phi^{\mathrm{w}}c^{\mathrm{F}}\nabla\varepsilon^{\mathrm{w}} + \beta\nabla\varepsilon^{+} - \gamma\nabla\varepsilon^{-})\cdot\nabla\theta\,\mathrm{d}\Omega = \int_{\Gamma_{J^\pm}}\theta(\mathbf{J}^{-*} - \mathbf{J}^{+*})\cdot\mathbf{n}\,\mathrm{d}\Gamma \tag{68}$$

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$$\int_\Omega s\partial(\phi^{\mathrm{w}}c^k)/\partial t\,\mathrm{d}\Omega + \int_\Omega (\xi\nabla\varepsilon^{\mathrm{w}} + \eta\nabla\varepsilon^+ + \zeta\nabla\varepsilon^-)\cdot\nabla s\,\mathrm{d}\Omega + \int_\Omega s\nabla\cdot(\phi^{\mathrm{w}}c^k\mathbf{v}^{\mathrm{s}})\,\mathrm{d}\Omega = -\int_{\Gamma_{J^\pm}} s(\mathbf{J}^{+*} + \mathbf{J}^{-*})\cdot\mathbf{n}\,\mathrm{d}\Gamma \tag{69}$$

where $\mathbf{t}^* = \boldsymbol{\sigma}^*\cdot\mathbf{n}$ is the external traction on the mixture boundary, and

$$\beta = \frac{\phi^{\mathrm{w}}c^+D^+}{\varepsilon^+} + k_D\frac{\phi^{\mathrm{w}}(c^+)^2}{\varepsilon^+} - k_D\frac{\phi^{\mathrm{w}}c^+c^-}{\varepsilon^+} \tag{70}$$

$$\gamma = \frac{\phi^{\mathrm{w}}c^-D^-}{\varepsilon^-} + k_D\frac{\phi^{\mathrm{w}}(c^-)^2}{\varepsilon^-} - k_D\frac{\phi^{\mathrm{w}}c^+c^-}{\varepsilon^-} \tag{71}$$

$$\xi = k_D\phi^{\mathrm{w}}c^k \tag{72}$$

$$\eta = \frac{\phi^{\mathrm{w}}c^+D^+}{\varepsilon^+} + k_D\frac{\phi^{\mathrm{w}}(c^+)^2}{\varepsilon^+} + k_D\frac{\phi^{\mathrm{w}}c^+c^-}{\varepsilon^+} \tag{73}$$

$$\zeta = \frac{\phi^{\mathrm{w}}c^-D^-}{\varepsilon^-} + k_D\frac{\phi^{\mathrm{w}}(c^-)^2}{\varepsilon^-} + k_D\frac{\phi^{\mathrm{w}}c^+c^-}{\varepsilon^-} \tag{74}$$

To solve this set of non-linear equations, the modified Newton–Raphson incremental iterative procedure is used. Let $\mathbf{u}^{\mathrm{s}} = \tilde{\mathbf{u}}^{\mathrm{s}} + \delta\mathbf{u}^{\mathrm{s}}$, $\varepsilon^{\mathrm{w}} = \tilde{\varepsilon}^{\mathrm{w}} + \delta\varepsilon^{\mathrm{w}}$, $\varepsilon^+ = \tilde{\varepsilon}^+ + \delta\varepsilon^+$, $\varepsilon^- = \tilde{\varepsilon}^- + \delta\varepsilon^-$ and other functions $f = \tilde{f} + \delta f$ (such as $\boldsymbol{\sigma} = \tilde{\boldsymbol{\sigma}} + \delta\boldsymbol{\sigma}$, $p = \tilde{p} + \delta p$, etc.), then the incremental equations are obtained,

$$\begin{aligned}
&\int_\Omega \{\mathrm{tr}[(\nabla\mathbf{w})^{\mathrm{T}}\cdot\tilde{\boldsymbol{\sigma}}^{\mathrm{e}}] - \tilde{p}\nabla\cdot\mathbf{w}\}\mathrm{d}\Omega + \int_\Omega \Bigg\{\mathrm{tr}[(\nabla\mathbf{w})^{\mathrm{T}}\cdot\delta\boldsymbol{\sigma}^{\mathrm{e}}] - \Bigg[\delta\varepsilon^{\mathrm{w}} - \left(\frac{\phi(c_0^{\mathrm{F}})^2}{\tilde{c}^k\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^3} + k_B\right)\nabla\cdot\delta\mathbf{u}^{\mathrm{s}} \\
&\quad + \frac{2\phi\tilde{\varepsilon}^+}{\tilde{c}^k\gamma_+\gamma_-}\delta\varepsilon^+ + \frac{2\phi\tilde{\varepsilon}^-}{\tilde{c}^k\gamma_+\gamma_-}\delta\varepsilon^-\Bigg]\nabla\cdot\mathbf{w}\Bigg\}\mathrm{d}\Omega = \int_{\Gamma_t}\mathbf{w}\cdot\mathbf{t}^*\,\mathrm{d}\Gamma
\end{aligned} \tag{75}$$

$$\begin{aligned}
&\int_\Omega g\nabla\cdot\mathbf{v}^{\mathrm{s}}\,\mathrm{d}\Omega + \int_\Omega k_D\left(\tilde{\phi}^{\mathrm{w}}\nabla\tilde{\varepsilon}^{\mathrm{w}} + \frac{\tilde{\phi}^{\mathrm{w}}\tilde{c}^+}{\tilde{\varepsilon}^+}\nabla\tilde{\varepsilon}^+ + \frac{\tilde{\phi}^{\mathrm{w}}\tilde{c}^-}{\tilde{\varepsilon}^-}\nabla\tilde{\varepsilon}^-\right)\cdot\nabla g\,\mathrm{d}\Omega \\
&\quad + \int_\Omega k_D\Bigg\{\tilde{\phi}^{\mathrm{w}}\nabla\delta\varepsilon^{\mathrm{w}} + \Bigg[\frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2}\nabla\tilde{\varepsilon}^{\mathrm{w}} + \frac{1}{\tilde{\varepsilon}^+}\left(\tilde{c}^+\nabla\tilde{\varepsilon}^+\frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2} - \frac{\tilde{\phi}^{\mathrm{w}}\nabla\tilde{\varepsilon}^+c_0^{\mathrm{F}}}{2\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\left(1+\frac{\tilde{c}^{\mathrm{F}}}{\tilde{c}^k}\right)\right) \\
&\quad + \frac{1}{\tilde{\varepsilon}^-}\left(\tilde{c}^-\nabla\tilde{\varepsilon}^-\frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2} + \frac{\tilde{\phi}^{\mathrm{w}}\nabla\tilde{\varepsilon}^-c_0^{\mathrm{F}}}{2\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\left(1-\frac{\tilde{c}^{\mathrm{F}}}{\tilde{c}^k}\right)\right)\Bigg]\nabla\cdot\delta\mathbf{u}^{\mathrm{s}} + \Bigg[\frac{1}{\tilde{\varepsilon}^+}\Bigg(\tilde{\phi}^{\mathrm{w}}\tilde{c}^+\nabla\delta\tilde{\varepsilon}^+ \\
&\quad - \frac{\tilde{\phi}^{\mathrm{w}}\tilde{c}^+}{\tilde{\varepsilon}^+}\nabla\tilde{\varepsilon}^+\delta\varepsilon^+ + \frac{\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^-}{\tilde{c}^k\gamma_+\gamma_-}\nabla\tilde{\varepsilon}^+\delta\varepsilon^+\Bigg) + \frac{\tilde{\phi}^{\mathrm{w}}}{\tilde{c}^k\gamma_+\gamma_-}\nabla\tilde{\varepsilon}^-\delta\varepsilon^+\Bigg] + \Bigg[\frac{1}{\tilde{\varepsilon}^-}\Bigg(\tilde{\phi}^{\mathrm{w}}\tilde{c}^-\nabla\delta\varepsilon^- \\
&\quad - \frac{\tilde{\phi}^{\mathrm{w}}\tilde{c}^-}{\tilde{\varepsilon}^-}\nabla\tilde{\varepsilon}^-\delta\varepsilon^- + \frac{\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^+}{\tilde{c}^k\gamma_+\gamma_-}\nabla\tilde{\varepsilon}^-\delta\varepsilon^-\Bigg) + \frac{\tilde{\phi}^{\mathrm{w}}}{\tilde{c}^k\gamma_+\gamma_-}\nabla\tilde{\varepsilon}^+\delta\varepsilon^-\Bigg]\Bigg\}\cdot\nabla g\,\mathrm{d}\Omega \\
&= -\int_{\Gamma_{J^{\mathrm{w}}}} g\mathbf{J}^{\mathrm{w}*}\cdot\mathbf{n}\,\mathrm{d}\Gamma
\end{aligned} \tag{76}$$

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$$\int_\Omega (k_D\tilde{\phi}^{\mathrm{w}}\tilde{c}^{\mathrm{F}}\nabla\varepsilon^{\mathrm{w}} + \tilde{\beta}\nabla\tilde{\varepsilon}^+ - \tilde{\gamma}\nabla\tilde{\varepsilon}^-)\cdot\nabla\theta\,\mathrm{d}\Omega + \int_\Omega \Bigg\{k_D\tilde{\phi}^{\mathrm{w}}\tilde{c}^{\mathrm{F}}\nabla\delta\varepsilon^{\mathrm{w}} + \Bigg[k_D\Bigg(\tilde{c}^{\mathrm{F}}\nabla\tilde{\varepsilon}^{\mathrm{w}}\frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2}$$

$$-\frac{\tilde{\phi}^{\mathrm{w}}\nabla\tilde{\varepsilon}^{\mathrm{w}}c_0^{\mathrm{F}}}{\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\Bigg) + \nabla\tilde{\varepsilon}^+\tilde{p}_1 - \nabla\tilde{\varepsilon}^-\tilde{p}_4\Bigg]\nabla\cdot\delta u^{\mathrm{s}} + [\nabla\tilde{\varepsilon}^+\tilde{p}_2\delta\varepsilon^+ + \tilde{\beta}\nabla\delta\varepsilon^+ - \nabla\tilde{\varepsilon}^-\tilde{p}_5\delta\varepsilon^+]$$

$$+ [\nabla\tilde{\varepsilon}^+\tilde{p}_3\delta\varepsilon^- - \tilde{\gamma}\nabla\delta\varepsilon^- - \nabla\tilde{\varepsilon}^-\tilde{p}_6\delta\varepsilon^-]\Bigg\}\cdot\nabla\theta\,\mathrm{d}\Omega = \int_{\Gamma_{J\pm}}\theta(\mathbf{J}^{-*} - \mathbf{J}^{+*})\cdot\mathbf{n}\,\mathrm{d}\Gamma \tag{77}$$

$$\int_\Omega s\Bigg\{\Bigg[\tilde{c}^k\frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2} - \frac{\tilde{\phi}^{\mathrm{w}}(c_0^{\mathrm{F}})^2}{\tilde{c}^k\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^3} + \tilde{\phi}^{\mathrm{w}}\tilde{c}^k\Bigg]\nabla\cdot\mathbf{v}^{\mathrm{s}} + \mathbf{v}^{\mathrm{s}}\cdot\nabla(\tilde{\phi}^{\mathrm{w}}\tilde{c}^k) + \frac{2\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^+}{\tilde{c}^k\gamma_+\gamma_-}\frac{\partial\varepsilon^-}{\partial t}$$

$$+\frac{2\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^-}{\tilde{c}^k\gamma_+\gamma_-}\frac{\partial\varepsilon^+}{\partial t}\Bigg\}\mathrm{d}\Omega + \int_\Omega(\tilde{\xi}\nabla\tilde{\varepsilon}^{\mathrm{w}} + \tilde{\eta}\nabla\tilde{\varepsilon}^+ + \tilde{\zeta}\nabla\tilde{\varepsilon}^-)\cdot\nabla s\,\mathrm{d}\Omega + \int_\Omega\Bigg\{k_D\tilde{\phi}^{\mathrm{w}}\tilde{c}^k\nabla\delta\varepsilon^{\mathrm{w}}$$

$$+\Bigg[k_D\nabla\tilde{\varepsilon}^{\mathrm{w}}\Bigg(\tilde{c}^k\frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2} - \frac{\tilde{\phi}^{\mathrm{w}}(c_0^{\mathrm{F}})^2}{\tilde{c}^k\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^3}\Bigg) + \nabla\tilde{\varepsilon}^+\tilde{p}_7 + \nabla\tilde{\varepsilon}^-\tilde{p}_{10}\Bigg]\nabla\cdot\delta\mathbf{u}^{\mathrm{s}}$$

$$+\Bigg(k_D\nabla\tilde{\varepsilon}^{\mathrm{w}}\frac{2\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^-}{\tilde{c}^k\gamma_+\gamma_-}\delta\varepsilon^+ + \nabla\tilde{\varepsilon}^+\tilde{p}_8\delta\varepsilon^+ + \tilde{\eta}\nabla\delta\varepsilon^+ + \nabla\tilde{\varepsilon}^-\tilde{p}_{11}\delta\varepsilon^+\Bigg) + \Bigg(k_D\nabla\tilde{\varepsilon}^{\mathrm{w}}\frac{2\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^+}{\tilde{c}^k\gamma_+\gamma_-}\delta\varepsilon^-$$

$$+\nabla\tilde{\varepsilon}^+\tilde{p}_9\delta\varepsilon^- + \tilde{\zeta}\nabla\delta\varepsilon^- + \nabla\tilde{\varepsilon}^-\tilde{p}_{12}\delta\varepsilon^-\Bigg)\Bigg\}\cdot\nabla s\,\mathrm{d}\Omega = -\int_{\Gamma_{J\pm}}s(\mathbf{J}^{+*} + \mathbf{J}^{-*})\cdot\mathbf{n}\,\mathrm{d}\Gamma \tag{78}$$

where $\tilde{f}$ represents the function value from last incremental iteration and δf represents the incremental value, and definitions of p_i $(i = 1, \ldots, 12)$ are in Appendix III. The non-dimensional parameter (k_B) is defined as $k_B = B_w/RTc_0$.

Now we apply the spatial discretization (finite element discretization) to the four primary DOFs, i.e., $\mathbf{u}^{\mathrm{s}}$, ε^{w}, ε^+ and ε^-, by using the standard Galerkin finite element procedure as follows:

$$\mathbf{u}^{\mathrm{s}} = \mathbf{N}\mathbf{u}_n, \quad \varepsilon^{\mathrm{w}} = \mathbf{H}\boldsymbol{\varepsilon}_n^{\mathrm{w}}, \quad \varepsilon^+ = \mathbf{H}\boldsymbol{\varepsilon}_n^-, \quad \varepsilon^- = \mathbf{H}\boldsymbol{\varepsilon}_n^-, \quad \boldsymbol{\varepsilon} = \mathbf{B}\mathbf{u}_n, \quad e = \mathbf{m}\boldsymbol{\varepsilon} = \mathbf{m}\mathbf{B}\mathbf{u}_n \tag{79}$$

where subscript n represents the nodal values, $\mathbf{N}$ and $\mathbf{H}$ are appropriate shape function matrices possessing C^0continuity in Ω. The C^0 continuity requirement is consistent with the highest order (first order) of derivatives of the primary variables in the formulation. Note that the same derivative requirement applies to $\mathbf{u}^{\mathrm{s}}$ as well as ε^{w}, $\varepsilon^+$and ε^-. Therefore, same C^0 continuity requirement is implied for all the primary variables. The choice of a particular shape function will depend on the specific problems to be solved and accuracy to be obtained. In this paper, we will solve only one dimensional (1 − D) examples where isoparametric linear shape functions will be used for all primary variables.

$\boldsymbol{\varepsilon} = [\varepsilon_{11}, \varepsilon_{22}, \varepsilon_{33}, 2\varepsilon_{13}, 2\varepsilon_{23}, 2\varepsilon_{12}]^{\mathrm{T}}$ is the strain vector of the solid matrix, e is the dilatation, $\mathbf{B}$ is the strain–displacement relationship in the standard finite element technique, and $\mathbf{m} = [1, 1, 1, 0, 0, 0]^{\mathrm{T}}$ is the Kronecker delta operator. Substituting equation (79) into equations (75)–(78), a matrix form of the finite element equations can be obtained,

$$\mathbf{C}^e\dot{\mathbf{d}}_n^e + \mathbf{A}^e\Delta\mathbf{d}_n^e + \mathbf{b}_n^e = \mathbf{F}_n^e \tag{80}$$

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where

$$\mathbf{d}_n^e = [\mathbf{u}_n,\ \varepsilon_n^{\mathrm{w}},\ \varepsilon_n^{+},\ \varepsilon_n^{-}]^{\mathrm{T}},\quad \mathbf{b}_n^e = [\mathbf{b}_{1n},\quad \mathbf{b}_{2n},\quad \mathbf{b}_{3n},\quad \mathbf{b}_{4n}]^{\mathrm{T}},\quad \mathbf{F}_n^e = [\mathbf{F}_{1n},\quad \mathbf{F}_{2n},\quad \mathbf{F}_{3n},\quad \mathbf{F}_{4n}]^{\mathrm{T}}$$

$$\mathbf{C}^e = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{C}_{2\mathrm{u}} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{C}_{4\mathrm{u}} & \mathbf{0} & \mathbf{C}_{4+} & \mathbf{C}_{4-} \end{bmatrix},\quad \mathbf{A}^e = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} & \mathbf{A}_{13} & \mathbf{A}_{14} \\ \mathbf{A}_{21} & \mathbf{A}_{22} & \mathbf{A}_{23} & \mathbf{A}_{24} \\ \mathbf{A}_{31} & \mathbf{A}_{32} & \mathbf{A}_{33} & \mathbf{A}_{34} \\ \mathbf{A}_{41} & \mathbf{A}_{42} & \mathbf{A}_{43} & \mathbf{A}_{44} \end{bmatrix}$$

$$\mathbf{D_1} = \mathbf{m}^{\mathrm{T}}\mathbf{m} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},\quad \mathbf{D_2} = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Xi = \lambda_{\mathrm{s}}\mathbf{D_1} + \mu_{\mathrm{s}}\mathbf{D_2},\quad \mathbf{Q} = \nabla\cdot\mathbf{N},\quad \Theta = \nabla\mathbf{H},$$

$$\mathbf{A}_{11} = \int_\Omega \left[\mathbf{B}^{\mathrm{T}}\Xi\mathbf{B} + \left(\frac{\phi(c_0^{\mathrm{F}})^2}{\tilde{c}^k\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^3} + k_{\mathrm{B}}\right)\mathbf{Q}^{\mathrm{T}}\mathbf{Q}\right]\mathrm{d}\Omega,\quad \mathbf{A_{12}} = -\int_\Omega \mathbf{Q}^{\mathrm{T}}\mathbf{H}\,\mathrm{d}\Omega$$

$$\mathbf{A}_{13} = -\int_\Omega \frac{2\phi\tilde{\varepsilon}^-}{\tilde{c}^k\gamma_+\gamma_-}\mathbf{Q}^{\mathrm{T}}\mathbf{H}\,\mathrm{d}\Omega,\quad \mathbf{A}_{14} = -\int_\Omega \frac{2\phi\tilde{\varepsilon}^+}{\tilde{c}^k\gamma_+\gamma_-}\mathbf{Q}^{\mathrm{T}}\mathbf{H}\,\mathrm{d}\Omega$$

$$\mathbf{b}_{1\mathrm{n}} = \int_\Omega (\mathbf{B}^{\mathrm{T}}\Xi\mathbf{B}\tilde{\mathbf{u}}_n - \tilde{p}\mathbf{Q}^{\mathrm{T}})\,\mathrm{d}\Omega,\quad \mathbf{F}_{1n} = \int_{\Gamma_t}\mathbf{N}^{\mathrm{T}}\mathbf{t}^*\,\mathrm{d}\Gamma$$

$$\mathbf{A}_{21} = \int_\Omega \Theta^{\mathrm{T}}k_D\left\{\frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2}\nabla\tilde{\varepsilon}^{\mathrm{w}} + \frac{1}{\tilde{\varepsilon}^+}\left[\tilde{c}^+\nabla\tilde{\varepsilon}^+\frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2} - \frac{\tilde{\phi}^{\mathrm{w}}\nabla\tilde{\varepsilon}^+c_0^{\mathrm{F}}}{2\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\left(1+\frac{\tilde{c}^{\mathrm{F}}}{\tilde{c}^k}\right)\right]\right.$$
$$\left. + \frac{1}{\tilde{\varepsilon}^-}\left(\tilde{c}^-\nabla\tilde{\varepsilon}^-\frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2} + \frac{\tilde{\phi}^{\mathrm{w}}\nabla\tilde{\varepsilon}^-c_0^{\mathrm{F}}}{2\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\left(1-\frac{\tilde{c}^{\mathrm{F}}}{\tilde{c}^k}\right)\right)\right\}\mathbf{Q}\,\mathrm{d}\Omega$$

$$\mathbf{A}_{22} = \int_\Omega Q^{\mathrm{T}}k_D\tilde{\phi}^{\mathrm{w}}Q\,\mathrm{d}\Gamma$$

$$\mathbf{A}_{23} = \int_\Omega \Theta^{\mathrm{T}}k_D\left[\frac{1}{\tilde{\varepsilon}^+}\left(\tilde{\phi}^{\mathrm{w}}\tilde{c}^+\Theta - \frac{\tilde{\phi}^{\mathrm{w}}\tilde{c}^+}{\tilde{\varepsilon}^+}\nabla\tilde{\varepsilon}^+\mathbf{H} + \frac{\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^-}{\tilde{c}^k\gamma_+\gamma_-}\nabla\tilde{\varepsilon}^+\mathbf{H}\right) + \frac{\tilde{\phi}^{\mathrm{w}}}{\tilde{c}^k\gamma_+\gamma_-}\nabla\tilde{\varepsilon}^-\mathbf{H}\right]\mathrm{d}\Omega$$

$$\mathbf{A}_{24} = \int_\Omega \Theta^{\mathrm{T}}k_D\left[\frac{1}{\tilde{\varepsilon}^-}\left(\tilde{\phi}^{\mathrm{w}}\tilde{c}^-\Theta - \frac{\tilde{\phi}^{\mathrm{w}}\tilde{c}^-}{\tilde{\varepsilon}^-}\nabla\tilde{\varepsilon}^-\mathbf{H} + \frac{\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^+}{\tilde{c}^k\gamma_+\gamma_-}\nabla\tilde{\varepsilon}^-\mathbf{H}\right) + \frac{\tilde{\phi}^{w}}{\tilde{c}^k\gamma_+\gamma_-}\nabla\tilde{\varepsilon}^+\mathbf{H}\right]\mathrm{d}\Omega$$

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$$\mathbf{C}_{2u} = \int_\Omega \mathbf{H}^{\mathrm{T}}\mathbf{Q}\,\mathrm{d}\Omega, \quad \mathbf{b}_{2n} = \int_\Omega \Theta^{\mathrm{T}} k_D\left(\tilde{\phi}^{w}\nabla\tilde{\varepsilon}^{\mathrm{w}} + + \frac{\tilde{\phi}^{\mathrm{w}}\tilde{c}^{+}}{\tilde{\varepsilon}^{+}}\nabla\tilde{\varepsilon}^{+} + \frac{\tilde{\phi}^{\mathrm{w}}\tilde{c}^{-}}{\tilde{\varepsilon}^{-}}\nabla\tilde{\varepsilon}^{-}\right)\mathrm{d}\Omega$$

$$\mathbf{F}_{2n} = -\int_{\Gamma_{J^{w}}} \mathbf{H}^{\mathrm{T}}\mathbf{J}^{\mathrm{w}*}\cdot\mathbf{n}\,\mathrm{d}\Gamma$$

$$\mathbf{A}_{31} = \int_\Omega \Theta^{\mathrm{T}}\left[k_D\left(\tilde{c}^{\mathrm{F}}\nabla\tilde{\varepsilon}^{\mathrm{w}}\frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2} - \frac{\tilde{\phi}^{\mathrm{w}}\nabla\tilde{\varepsilon}^{\mathrm{w}}c_0^{\mathrm{F}}}{\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\right) + \nabla\tilde{\varepsilon}^{+}\tilde{p}_1 - \nabla\tilde{\varepsilon}^{-}\tilde{p}_4\right]\mathbf{Q}\,\mathrm{d}\Omega$$

$$\mathbf{A}_{32} = \int_\Omega \Theta^{\mathrm{T}} k_D\tilde{\phi}^{\mathrm{w}}\tilde{c}^{\mathrm{F}}\Theta\,\mathrm{d}\Omega, \quad \mathbf{A}_{33} = \int_\Omega \Theta^{\mathrm{T}}\left[\nabla\tilde{\varepsilon}^{+}\tilde{p}_2\mathbf{H} + \tilde{\beta}\Theta - \nabla\tilde{\varepsilon}^{-}\tilde{p}_5\mathbf{H}\right]\mathrm{d}\Omega$$

$$\mathbf{A}_{34} = \int_\Omega \Theta^{\mathrm{T}}\left[\nabla\tilde{\varepsilon}^{+}\tilde{p}_3\mathbf{H} - \tilde{\gamma}\Theta - \nabla\tilde{\varepsilon}^{-}\tilde{p}_6\mathbf{H}\right]\mathrm{d}\Omega$$

$$\mathbf{b}_{3n} = \int_\Omega \Theta^{\mathrm{T}}(k_D\tilde{\phi}^{\mathrm{w}}\tilde{c}^{\mathrm{F}}\nabla\tilde{\varepsilon}^{\mathrm{w}} + \tilde{\beta}\nabla\tilde{\varepsilon}^{+} - \tilde{\gamma}\nabla\tilde{\varepsilon}^{-})\,\mathrm{d}\Omega, \quad \mathbf{F}_{3n} = \int_{\Gamma_{J\pm}} (\mathbf{J}^{-*} - \mathbf{J}^{+*})\cdot\mathbf{n}\,\mathrm{d}\Gamma$$

$$\mathbf{A}_{41} = \int_\Omega \Theta^{\mathrm{T}}\left[k_D\nabla\tilde{\varepsilon}^{\mathrm{w}}\left(\tilde{c}^{k}\frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2} - \frac{\tilde{\phi}^{\mathrm{w}}(c_0^{\mathrm{F}})^2}{\tilde{c}^{k}\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^3}\right) + \nabla\tilde{\varepsilon}^{+}\tilde{p}_7 + \nabla\tilde{\varepsilon}^{-}\tilde{p}_{10}\right]\mathbf{Q}\,\mathrm{d}\Omega$$

$$\mathbf{A}_{42} = \int_\Omega \Theta^{\mathrm{T}} k_D\tilde{\phi}^{\mathrm{w}}\tilde{c}^{k}\Theta\,d\Omega$$

$$\mathbf{A}_{43} = \int_\Omega \Theta^{\mathrm{T}}\left[k_D\nabla\tilde{\varepsilon}^{\mathrm{w}}\frac{2\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^{-}}{\tilde{c}^{k}\gamma_{+}\gamma_{-}}\mathbf{H} + \nabla\tilde{\varepsilon}^{+}\tilde{p}_8\mathbf{H} + \tilde{\eta}\Theta + \nabla\tilde{\varepsilon}^{-}\tilde{p}_{11}\mathbf{H}\right]\mathrm{d}\Omega$$

$$\mathbf{A}_{44} = \int_\Omega \Theta^{\mathrm{T}}\left[k_D\nabla\tilde{\varepsilon}^{\mathrm{w}}\frac{2\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^{+}}{\tilde{c}^{k}\gamma_{+}\gamma_{-}}\mathbf{H} + \nabla\tilde{\varepsilon}^{+}\tilde{p}_9\mathbf{H} + \tilde{\zeta}\Theta + \nabla\tilde{\varepsilon}^{-}\tilde{p}_{12}\mathbf{H}\right]\mathrm{d}\Omega$$

$$\mathbf{C}_{4\mathrm{u}} = \int_\Omega \mathbf{H}^{\mathrm{T}}\left\{\left[\tilde{c}^{k}\frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2} - \frac{\tilde{\phi}^{\mathrm{w}}(c_0^{\mathrm{F}})^2}{\tilde{c}^{k}\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^3} + \tilde{\phi}^{\mathrm{w}}\tilde{c}^{k}\right]\mathbf{Q} + \nabla^{\mathrm{T}}(\tilde{\phi}^{\mathrm{w}}\tilde{c}^{k})\mathbf{N}\right\}\mathrm{d}\Omega$$

$$\mathbf{C}_{4+} = \int_\Omega \mathbf{H}^{\mathrm{T}}\frac{2\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^{+}}{\tilde{c}^{k}\gamma_{+}\gamma_{-}}\mathbf{H}\,\mathrm{d}\Omega, \quad \mathbf{C}_{4-} = \int_\Omega \mathbf{H}^{\mathrm{T}}\frac{2\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^{-}}{\tilde{c}^{k}\gamma_{+}\gamma_{-}}\mathbf{H}\mathrm{d}\Omega$$

$$\mathbf{b}_{4n} = \int_\Omega \Theta^{\mathrm{T}}(\tilde{\xi}\nabla\tilde{\varepsilon}^{\mathrm{w}} + \tilde{\eta}\nabla\tilde{\varepsilon}^{+} + \tilde{\zeta}\nabla\tilde{\varepsilon}^{-})\,\mathrm{d}\Omega, \quad \mathbf{F}_{4n} = -\int_{\Gamma_{J\pm}} \mathbf{H}^{\mathrm{T}}(\mathbf{J}^{+*} + \mathbf{J}^{-*})\cdot\mathbf{n}\,\mathrm{d}\Gamma \tag{81}$$

It should be noted that the current formulation is similar to the mixed u–p finite element formulation for the saturated porous media (for biphasic mixture) in terms of treatment of the pressure term [55, 56, 25, 26]. The main difference is that this paper deals with a new triphasic mechano-electrochemical theory in which the ion concentrations and electrical portential are considered as well as the solid deformation and fluid pressure. If the ion concentration and

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electrical potential effects are neglected, the triphasic theory reduces to the biphasic theory [9], the water chemical potential essentially reduces to fluid pressure and the current triphasic finite element formulation reduces to the biphasic u–p formulation. In addition, our formulation involves the first-order derivatives of the chemical potential for water ε^{w}, electrochemical potentials for cation ε^{+} and anion ε^{-} as well as the solid displacement $\mathbf{u}^{\mathrm{s}}$. The same highest-order derivatives of all primary DOFs lead to the same continuity requirement on them; therefore, we do not require different order shape functions for the quantities $\mathbf{u}^{\mathrm{s}}$, ε^{w}, ε^{+} and ε^{-}.

Standard assembly operations can be performed to sum up the element contributions, which produces a first-order ordinary differential equation with respect to time:

$$\mathbf{C}\dot{\mathbf{d}}_n + \mathbf{A}\Delta\,\mathbf{d}_n + \mathbf{b}_n = \mathbf{F}_n \tag{82}$$

The first-order differential equations can be solved using the finite difference method. The implicit Euler backward scheme is used from the consideration of its unconditional stability,

$$^{t+\Delta t}\dot{\mathbf{d}}_n^{(i)} = \frac{^{t+\Delta t}\mathbf{d}_n^{(i-1)} + \Delta\mathbf{d}_n^{(i)} - {}^{t}\mathbf{d}_n}{\Delta t} \tag{83}$$

where i is the index of iteration within each time step. Then the final algebraic equation system is

$$\left(\frac{1}{\Delta t}\,{}^{t}\mathbf{C} + {}^{t}\mathbf{A}\right)\Delta\mathbf{d}_n^{(i)} = {}^{t+\Delta t}\mathbf{F}_n - {}^{t+\Delta t}\mathbf{b}_n^{(i-1)} - {}^{t+\Delta t}\mathbf{C}^{(i-1)}\frac{^{t+\Delta t}\mathbf{d}_n^{(i-1)} - {}^{t}\mathbf{d}_n}{\Delta t} \tag{84}$$

Within each time step, non-linear iterations are repeated until the incremental values $\Delta\mathbf{d}_n^{(i)}$ are small enough (less than the tolerance), i.e. the convergence criteria is

$$|\Delta\mathbf{d}_n^{(i)}| < \text{tolerance} \tag{85}$$

EXAMPLARY PROBLEMS

The finite element formulation is implemented for one dimensional (1-D) cases using 2-node linear isoparametric element for solid displacement field and modified electrochemical/chemical potentials. A 1-D free swelling problem (i.e. no applied tractions) and triphasic stress relaxation (i.e. displacement-controlled) problem were investigated using this FEM formulation. The FEM results were checked and compared with those obtained by the finite difference method (FDM).

Free swelling problem

To investigate the mechano-electrochemical coupling phenomena of charged, hydrated soft tissues, a transient free swelling problem was studied using our FEM (Figure 2(a)). The shape functions N and H used in this one-dimensional problem are isoparametric linear functions as shown in Figure 2(a). Assuming a frictionless lateral boundary, only axial motion is possible and the problem is reduced to a 1-D problem. Initially, a tissue sample was at equilibrium with the external bathing solution with concentration c_0^* of mono-valent ions (e.g. Na^+ and Cl^-). The

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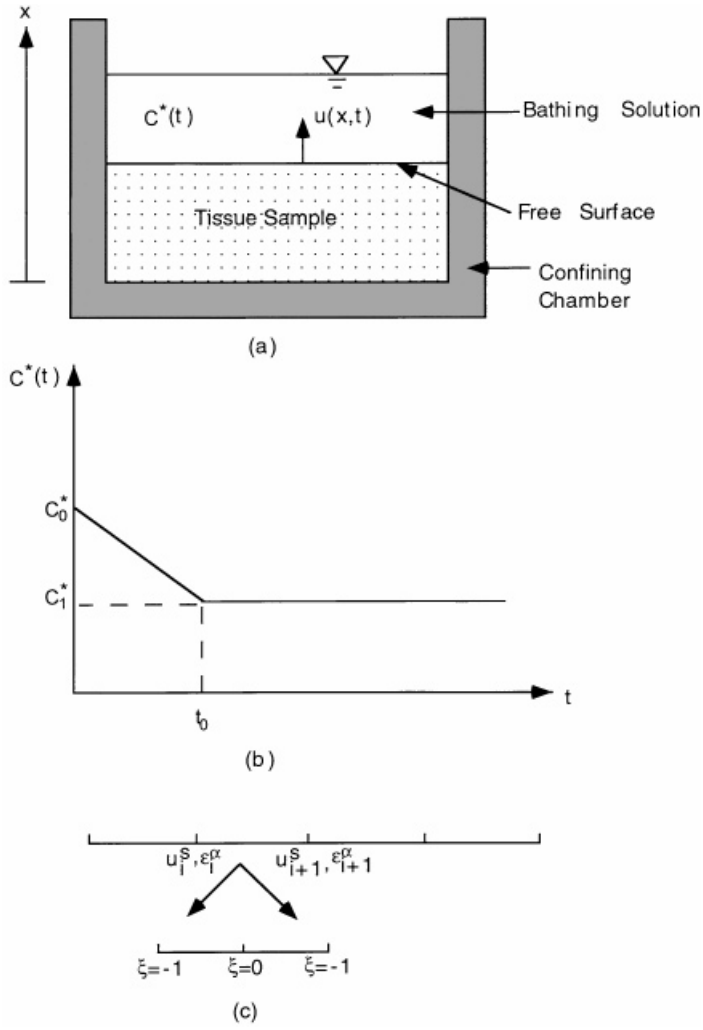

Figure 2. Schematic representation of the transient free swelling problem: (a) Sample of charged-hydrated tissue of thickness h is bathed in the monovalent solution; (b) the external solution concentrations changes from c_0^* to c_1^* linearly within t_0; (c) 2-node linear isoparametric element

lateral sides of the chamber are impermeable and confine the lateral movement of the tissue. The boundary conditions at the bottom of the specimen are that solid displacement and all fluxes are zero. The boundary conditions at the top are that electrochemical/chemical potentials and total stress are continuous across the boundary. At $t = 0^+$, the concentration of external solution at top decreases to c_1^* linearly (Figure 2(b)). This lowering of c^* from c_0^* to c_1^* causes the tissue to swell to a new equilibrium state. Figure 2(c) shows the schematic mesh of isoparametric elements. The transient responses of the solid displacement, ion concentrations, fluid and ion velocities to the chemical loading (change of external ion concentration) are solved using our FEM and compared with those obtained by the FDM technique.

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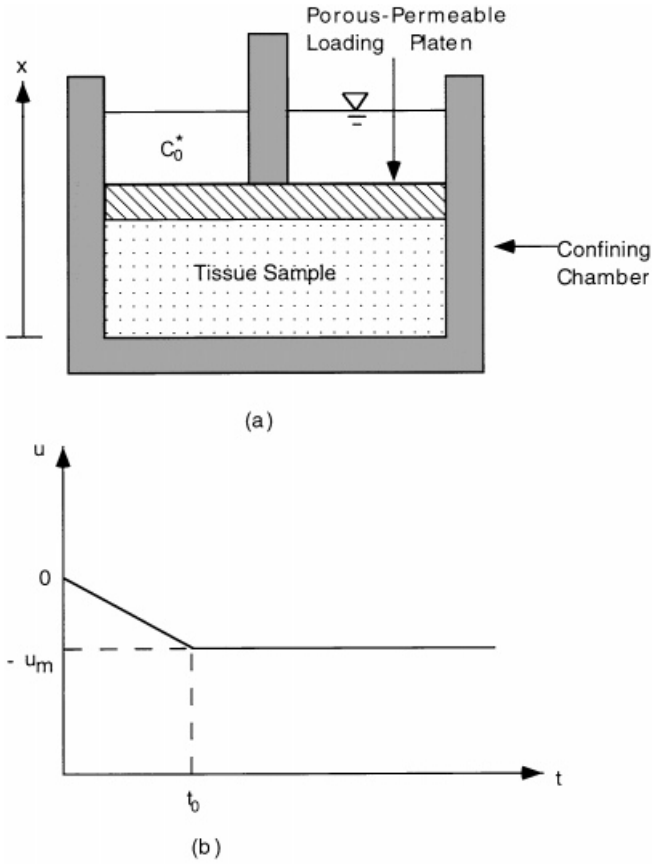

Figure 3. Schematic representation of the stress relaxation test on the hydrated soft tissue in confined compression configuration: (a) The sample with the thickness h bathed in the monovalent solution is subjected to a displacement loading at the top surface via a rigid porous permeable platen; (b) a ramp displacement is applied during $0 < t < t_0$. The maximum surface-to-surface compressive strain applied at the top is 10 per cent

Stress relaxation problem

Figure 3 is a schematic diagram of the confined compression stress relaxation experiment. A soft tissue with a thickness h is placed in a confining chamber (Figure 3(a)). The shape functions **N** and **H** used in this one-dimensional problem are similar to the ones used above. The boundary conditions at bottom are the same as those in the free swelling problem mentioned above. In this problem, however, at the top surface ($x = h$) a ramp displacement (Figure 3(b)) is applied through a rigid porous-permeable (i.e. free draining) loading platen. The electrochemical/chemical potentials are continuous across this boundary. The concentration c_0^* of the external solution is kept constant during the test (i.e. neglecting the effect of ion transfer). The solid displacement, ion concentrations, fluid and ion fluxes within the tissue, and the resultant stress are calculated using our FEM. Note that the initial equilibrium free swelling state is used as the reference configuration for both free swelling and stress relaxation problems.

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RESULTS

Free swelling problem

A total of 50 uniform elements were used for this problem. The material parameters used for FEM calculation were $h = 0{\cdot}5$ mm, $\lambda_s + 2\mu_s = 0{\cdot}5$ MPa, $B_w = 0$, $D^+ = 0{\cdot}5 \times 10^{-9}$ m^2/s, $D^- = 0{\cdot}8 \times 10^{-9}$ m^2/s, $\alpha = 0{\cdot}7 \times 10^{15}$ N s/m^4, $T = 298$ K, $\phi_0^w = 0{\cdot}75$, and $c_0^F = 0{\cdot}20$ mEq/ml. The basic scales for non-dimensionalization were taken as $c_0 = 0{\cdot}1$M and $D_0 = D^+$. Initially, the external solution concentration was 0·15 M. It decreased linearly to 0·125 M within 10 s and then remained constant (Figure 2(b)). The time step was 1 s. Immediately after dilution of the external solution, the tissue begins to swell. Figure 4 shows the history of the solid displacement at the surface. This swelling result is consistent with our experimental observation and those of other investigations [57]. To check the accuracy of our FEM calculation, the result was compared with that obtained using FDM with 101 grid points. As shown in Figure 4, the agreement between two calculations is excellent ($[\max(|\mathbf{u}_{FDM} - \mathbf{u}_{FEM}|/\mathbf{u}_{FDM})] < 0{\cdot}1$per cent).

Figure 5 shows the cation concentration distributions within the tissues during the free swelling process. Again, the FEM results are in excellent agreement with those calculated by FDM. To investigate the effects of the FCD and ion diffusivities on the free swelling behavior of charged hydrated soft tissues, we calculated the solid displacement at surface as a function of FCD (Figure 6) as well as a function of the ratio of ion diffusivities (Figure 7). As shown in Figure 6, the equilibrium swelling strain increases as the FCD increases. At early time an interesting anomalous displacement in the tissue swelling is noted as the FCD is decreased. That is, for certain values of FCD (e.g. FCD = 0·05 mEq/ml), the tissue shrinks initially (< 210 s). Figure 7 shows the surface displacement history as a function of D^+/D^- (where $D^- = 0{\cdot}8 \times 10^{-9}$ m^2/s is fixed). As shown in Figure 7, the increase of D^+/D^- facilitates tissue swelling. Interestingly, when $D^+/D^- > 1$, instead of swelling monotonically, there is an 'overshoot' in the tissue swelling, but finally the tissue reaches the same equilibrium state as for the case where $D^+/D^- \leqslant 1$. These anomalous phenomena have underlying physical mechanisms and are believed to be related to the negative osmosis effects [47]. The detailed analysis will be presented in a separate paper.

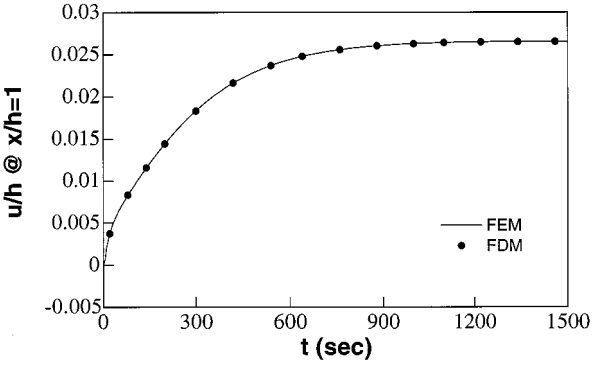

Figure 4. The comparison of the surface displacements in the free swelling test from finite element and finite difference calculations. In FEM, 50 uniform element mesh was used; in FDM, 101 uniform grid points were used

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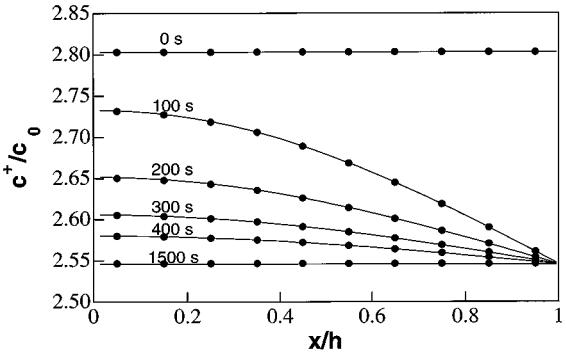

Figure 5. The comparison of the cation concentration distributions in the free swelling test from FEM (solid lines) and FDM calculations (dots). The normalization scale $c_0 = 0{\cdot}1$ M

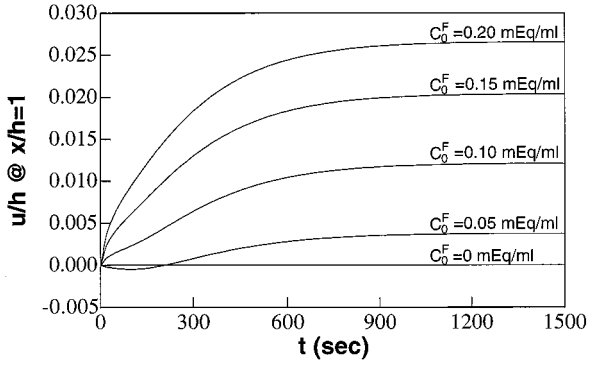

Figure 6. Free swelling history as a function of tissue FCD

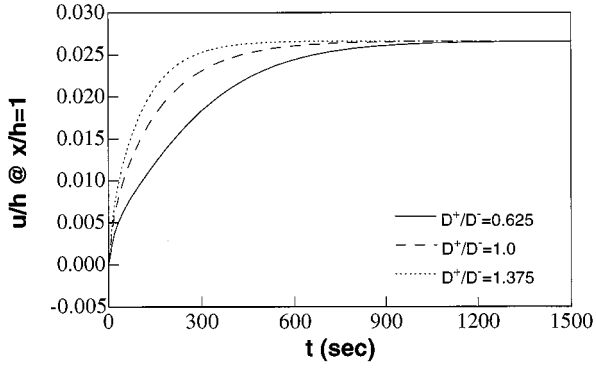

Figure 7. The surface displacement history with D^+/D^- as the parameter in the free swelling test. D^- is fixed

Stress relaxation problem

A total of 50 uniform elements were used for this problem. The material properties used for the FEM calculation were identical to those used in the free swelling problem except

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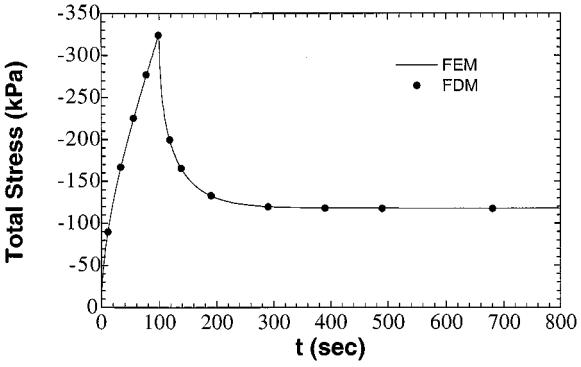

Figure 8. The comparison of the total stress histories in the triphasic stress relaxation test between FEM calculation with 50 elements used and FDM result with 101 grid points used

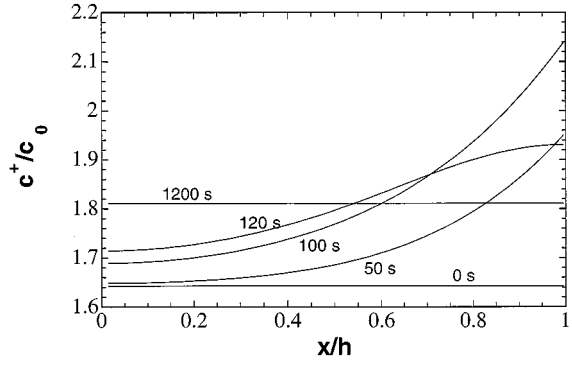

Figure 9. The prediction of the cation distribution within the tissue at different times in the triphasic stress relaxation test

$c_0^F = 0{\cdot}26$ mEq/ml. The external bath concentration c_0^* is 0·15 M NaCl. A 10 per cent surface-to-surface compressive strain in $t_0 = 100$ s was applied to the surface and the time step was chosen as 0·1 s. The basic scales were taken as $c_0 = 0{\cdot}2$ M and $D_0 = D^+$.

Figure 8 shows the stress relaxation history of the charged hydrated soft tissue. This result was again compared with that calculated using the FDM with a total of 101 grids. As shown in Figure 8, there is an excellent agreement between the two methods; such stress-relaxation behaviors have long been reported in the literature (e.g. [4, 18, 19, 51]).

Figure 9 shows the distributions of c^+ within the tissues during stress relaxation, calculated by the FEM. It should be mentioned that all FEM results are in excellent agreement with those by the FDM [51]. The counter-ion concentration c^+ increases towards the tissue surface because of the increase of c^F due to higher compaction near the surface (see equation (12)). The distribution of ion concentration finally reaches uniform at equilibrium at a higher level as predicted by equation (12).

Using the peak stress (Figure 8) as the indicator, the convergence of the finite element analysis was examined by refinement of mesh and the result is shown in Figure 10 and Table II. It can be

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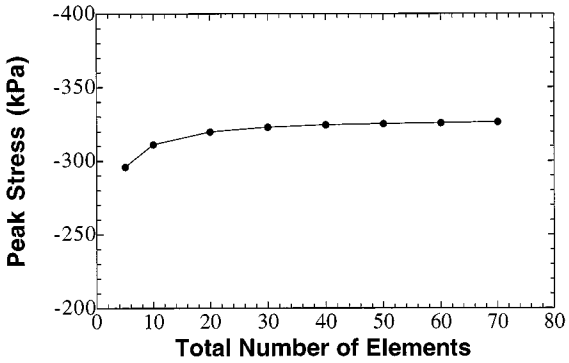

Figure 10. The mesh convergence with the peak total stress as the indicator. When over 30 elements are used, the peak stress has a variation of less than 0.5 per cent. See also Table II

Table II. The relative error of the peak total stress versus the number of elements

i	Element numbers	$\lvert\sigma_{i+1} - \sigma_i\rvert / \lvert\sigma_i\rvert$ (%)
1	5	5·20
2	10	2·76
3	20	0·94
4	30	0·48
5	40	0·29
6	50	0·19
7	60	0·14

seen that when the mesh incorporates more than 30 elements, the relative error of peak stress is less than 0·5 per cent. Therefore, a good mesh convergence has been achieved with more than 30 elements.

SUMMARY AND CONCLUSION REMARKS

In this study we developed a new equivalent expression of the triphasic mechano-electrochemical theory [9]. For this equivalent form, we derived a mixed finite element using the standard Galerkin weighted residual method. In this FEM formulation, solid displacement $\mathbf{u}^s$, modified electrochemical/chemical potentials ε^w, ε^+ and ε^- for water, cation and anion are the four primary DOFs and are independently interpolated. The modified Newton–Raphson iterative procedure was employed to handle the nonlinear terms. The resulting first order ODE with respect to time was solved using the implicit Euler backward scheme that is unconditionally stable. Since the system is not self-adjoint, the final algebraic equation system is not symmetric but it is still sparse. A 1-D linear isoparametric element was developed and the formulation accuracy and convergence have been tested with two illustrative examples.

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With the current choice of primary DOFs, the formulation has the advantage of small amount of storage (only one vector and three scalar variables). At each node there are only four unknowns in 1-D cases and there are only six unknowns for 3-D cases. Also, the jump conditions between elements and across the interface boundary are automatically satisfied. Moreover, the current formulation does not require special treatments (e.g. the penalty method) for this intrinsically incompressible system.

The 1-D free swelling and stress relaxation problems were studied. The solid displacement, ion concentrations, fluid and ion velocities were calculated using our FEM and compared with those using FDM. The results show that there is an excellent agreement between two methods. The numerical results are also consistent with many of the experimental observations on charged hydrated soft tissues reported in the literature [4, 19, 57]. No locking and spurious modes such as checkerboard phenomenen are found in the calculation. Through these sample cases, it is shown that our FEM formulation is capable of solving the complex triphasic problems of charged hydrated soft tissues under different types of mechanical, electrical and physicochemical loading conditions. With the success of 1-D cases, the multi-dimensional problems will be investigated in the future.

APPENDIX I

All the unknowns of the triphasic theory can be expressed with the four primary DOFs $\mathbf{u}^{\mathrm{s}}$, ε^{w}, ε^{+} and ε^{-}. The strain tensor $\mathbf{E}$ and the dilatation e can be obtained directly from $\mathbf{u}^{\mathrm{s}}$, and ϕ^{s} and c^{F} can be obtained from equations (9) and (10), respectively. The volume fraction of ions ϕ^{+} and ϕ^{-} are small and thus neglected, so the porosity can be approximated as $\phi^{\mathrm{w}} = 1 - \phi^{\mathrm{s}}$. Multiplying (25) with (26), we get

$$\varepsilon^{+}\varepsilon^{-} = \gamma_{+}\gamma_{-}c^{+}c^{-} \tag{86}$$

Substituting (86) into (10), we obtain

$$c^{+} = \frac{c^{\mathrm{F}} + \sqrt{(c^{\mathrm{F}})^{2} + \dfrac{4\varepsilon^{+}\varepsilon^{-}}{\gamma_{+}\gamma_{-}}}}{2} \tag{87}$$

$$c^{-} = \frac{-c^{\mathrm{F}} + \sqrt{(c^{\mathrm{F}})^{2} + \dfrac{4\varepsilon^{+}\varepsilon^{-}}{\gamma_{+}\gamma_{-}}}}{2} \tag{88}$$

Defining

$$c^{k} = c^{+} + c^{-} \tag{89}$$

then

$$c^{k} = \sqrt{(c^{\mathrm{F}})^{2} + \frac{4\varepsilon^{+}\varepsilon^{-}}{\gamma_{+}\gamma_{-}}} \tag{90}$$

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From equation (24) and (89), the fluid pressure can be expressed as

$$p = RT\varepsilon^{\mathrm{w}} + RT\phi c^{k} - B_{w}e \tag{91}$$

and with equation (25) and (26), the electrical potential is

$$\psi = \frac{RT}{F_{\mathrm{c}}}\ln\left(\frac{\varepsilon^{+}}{\gamma_{+}c^{+}}\right) \tag{92}$$

$$= \frac{-RT}{F_{\mathrm{c}}}\ln\left(\frac{\varepsilon^{-}}{\gamma_{-}c^{-}}\right) \tag{93}$$

$$= \frac{RT}{2F_{\mathrm{c}}}\ln\left(\frac{\gamma_{-}c^{-}\varepsilon^{+}}{\gamma_{+}c^{+}\varepsilon^{-}}\right) \tag{94}$$

APPENDIX II

Consider a material volume $R(t)$ of a triphasic mixture, whose boundary $\partial R(t)$ and surface of discontinuity $\Gamma(t)$ are material surfaces fixed on the solid phase. If there is no mass production and the solid phases on the two sides of the surface of discontinuity are not allowed to slide on each other, then

$$[[\mathbf{v}_{\mathrm{s}}]] = 0 \tag{95}$$

From the mass conservation of individual components, with the help of Reynolds transport theorem and equations (5) and (6), the following jump conditions on $\Gamma(t)$ can be easily derived:

$$[[\phi^{\mathrm{w}}(\mathbf{v}^{\mathrm{w}} - \mathbf{v}^{\mathrm{s}})]] = 0 \tag{96}$$

$$[[\phi^{\mathrm{w}}c^{+}(\mathbf{v}^{+} - \mathbf{v}^{\mathrm{s}})]] = 0 \tag{97}$$

$$[[\phi^{\mathrm{w}}c^{-}(\mathbf{v}^{-} - \mathbf{v}^{\mathrm{s}})]] = 0 \tag{98}$$

According to equations (27)–(29), (96)–(98) are the continuity of the volume fluxes

$$[[J^{\beta}]] = 0, \quad \beta = w, +, - \tag{99}$$

Similarly, if we assume the sum of the momentum supply to individual phases, the sum of the energy supply to individual phases, all of which come from the interaction between phases, are zero, then from the conservation of linear momentum and energy, and from entropy inequality, we have the jump conditions for the mixture,

$$\left[\left[\sum_{\alpha}\boldsymbol{\sigma}^{\alpha} - \sum_{\beta}\rho^{\beta}\mathbf{v}^{\beta}(\mathbf{v}^{\beta} - \mathbf{v}^{\mathrm{s}})\right]\right]\cdot\mathbf{n} = 0, \tag{100}$$

$$\left[\left[\sum_{\alpha}\boldsymbol{\sigma}^{\alpha}\cdot\mathbf{v}^{\alpha} - \sum_{\alpha}\mathbf{q}^{\alpha} - \sum_{\beta}\rho^{\beta}\left(e^{\beta} + \frac{1}{2}\mathbf{v}^{\beta}\cdot\mathbf{v}^{\beta}\right)(\mathbf{v}^{\beta} - \mathbf{v}^{\mathrm{s}})\right]\right]\cdot\mathbf{n} = 0 \tag{101}$$

$$\left[\left[\frac{1}{\theta}\sum_{\alpha}\mathbf{q}^{\alpha} + \sum_{\beta}\rho^{\beta}\eta^{\beta}(\mathbf{v}^{\beta} - \mathbf{v}^{\mathrm{s}})\right]\right]\cdot\mathbf{n} = 0 \tag{102}$$

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where $\alpha = \mathrm{s}, \mathrm{w}, +, -$, and $\beta = \mathrm{w}, +, -$. e^β is the internal energy of the β component. $\mathbf{q}^\alpha$ is the heat flux to the α component, η^α is the entropy of α component, and θ is the absolute temperature.

The (electro)chemical potentials are defined as

$$\mathbf{K}^{\mathrm{s}} = A^{\mathrm{s}}\mathbf{I} - \boldsymbol{\sigma}^{\mathrm{s}}/\rho^{\mathrm{s}} \tag{103}$$

$$\mathbf{K}^{\beta} = \tilde{\mu}^{\beta}\mathbf{I} = A^{\beta}\mathbf{I} - \boldsymbol{\sigma}^{\beta}/\rho^{\beta} \tag{104}$$

where $A^\alpha = e^\alpha - \theta\eta^\alpha$ is the Helmholz free energy of α component.

By neglecting the terms with velocity product, (100) gives

$$[[\boldsymbol{\sigma}]]\cdot\mathbf{n} = 0 \tag{105}$$

where

$$\boldsymbol{\sigma} = \sum_{\alpha}\boldsymbol{\sigma}^{\alpha} \tag{106}$$

For isothermal process, also neglecting velocity product, we obtain from (100)–(105),

$$\left[\left[\sum_{\beta}\rho^{\beta}\tilde{\mu}^{\beta}(\mathbf{v}^{\beta} - \mathbf{v}^{\mathrm{s}})\right]\right]\cdot\mathbf{n} = 0 \tag{107}$$

$\tilde{\mu}^\beta$'s are constitutively independent of the relative velocities $\mathbf{v}^\beta - \mathbf{v}^{\mathrm{s}}$, also $\mathbf{v}^\beta - \mathbf{v}^{\mathrm{s}}$ and ρ^β are independent variables [9, 36], so that $\tilde{\mu}^\beta$ is independent of $\rho^\beta(\mathbf{v}^\beta - \mathbf{v}^{\mathrm{s}})$, thus from (107) and (96)–(98) we obtain,

$$[[\mu^{\mathrm{w}}]] = 0 \tag{108}$$

$$[[\tilde{\mu}^{+}]] = 0 \tag{109}$$

$$[[\tilde{\mu}^{-}]] = 0 \tag{110}$$

APPENDIX III

Some intermediate quantities used in the finite element formulation are as follows:

$$\tilde{p}_1 = \frac{D^+}{\tilde{\varepsilon}^+}\left[\tilde{c}^+\frac{\phi_0^s}{(1+\tilde{e})^2} - \frac{\tilde{\phi}^{\mathrm{w}}c_0^{\mathrm{F}}}{2\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\left(1+\frac{\tilde{c}^{\mathrm{F}}}{\tilde{c}^k}\right)\right] + \frac{k_D}{\tilde{\varepsilon}^+}\left[(\tilde{c}^+)^2\frac{\phi_0^s}{(1+\tilde{e})^2}\right.$$
$$\left. - \frac{\tilde{\phi}^{\mathrm{w}}\tilde{c}^+c_0^{\mathrm{F}}}{\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\left(1+\frac{\tilde{c}^{\mathrm{F}}}{\tilde{c}^k}\right)\right] - \frac{k_D}{\tilde{\varepsilon}^+}\tilde{c}^+\tilde{c}^-\frac{\phi_0^s}{(1+\tilde{e})^2}$$

$$\tilde{p}_2 = \frac{D^+}{\tilde{\varepsilon}^+}\left(\frac{\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^-}{\tilde{c}^k\gamma_+\gamma_-} - \frac{\tilde{\phi}^{\mathrm{w}}\tilde{c}^+}{\tilde{\varepsilon}^+}\right) + \frac{k_D}{\tilde{\varepsilon}^+}\left[\frac{2\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^-\tilde{c}^+}{\tilde{c}^k\gamma_+\gamma_-} - \frac{\tilde{\phi}^{\mathrm{w}}(\tilde{c}^+)^2}{\tilde{\varepsilon}^+}\right] - \frac{k_D}{\tilde{\varepsilon}^+}\left(\frac{\tilde{\phi}^{\mathrm{w}}\tilde{\varepsilon}^-}{\gamma_+\gamma_-} - \frac{\tilde{\phi}^{\mathrm{w}}}{\tilde{\varepsilon}^+}\tilde{c}^+\tilde{c}^-\right)$$

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$$\tilde{p}_3 = \frac{\tilde{\phi}^{\mathrm{w}} D^+}{\tilde{c}^k \gamma_+ \gamma_-} + k_D \frac{2\tilde{\phi}^{\mathrm{w}} \tilde{c}^+}{\tilde{c}^k \gamma_+ \gamma_-} - k_D \frac{\tilde{\phi}^{\mathrm{w}}}{\gamma_+ \gamma_-}$$

$$\tilde{p}_4 = \frac{D^-}{\tilde{\varepsilon}^-}\left[\tilde{c}^- \frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2} + \frac{\tilde{\phi}^{\mathrm{w}} c_0^{\mathrm{F}}}{2\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\left(1 - \frac{\tilde{c}^{\mathrm{F}}}{\tilde{c}^k}\right)\right] + \frac{k_D}{\tilde{\varepsilon}^-}\left[(\tilde{c}^-)^2 \frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2}\right.$$
$$\left. + \frac{\tilde{\phi}^{\mathrm{w}} \tilde{c}^- c_0^{\mathrm{F}}}{\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\left(1 - \frac{\tilde{c}^{\mathrm{F}}}{\tilde{c}^k}\right)\right] - \frac{k_D}{\tilde{\varepsilon}^-}\tilde{c}^+ \tilde{c}^- \frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2}$$

$$\tilde{p}_5 = \frac{\tilde{\phi}^{\mathrm{w}} D^-}{\tilde{c}^k \gamma_+ \gamma_-} + k_D \frac{2\tilde{\phi}^{\mathrm{w}} \tilde{c}^-}{\tilde{c}^k \gamma_+ \gamma_-} - k_D \frac{\tilde{\phi}^{\mathrm{w}}}{\gamma_+ \gamma_-}$$

$$\tilde{p}_6 = \frac{D^-}{\tilde{\varepsilon}^-}\left(\frac{\tilde{\phi}^{\mathrm{w}} \tilde{\varepsilon}^+}{\tilde{c}^k \gamma_+ \gamma_-} - \frac{\tilde{\phi}^{\mathrm{w}} \tilde{c}^-}{\tilde{\varepsilon}^-}\right) + \frac{k_D}{\tilde{\varepsilon}^-}\left[\frac{2\tilde{\phi}^{\mathrm{w}} \tilde{\varepsilon}^+ \tilde{c}^-}{\tilde{c}^k \gamma_+ \gamma_-} - \frac{\tilde{\phi}^{\mathrm{w}} (\tilde{c}^-)^2}{\tilde{\varepsilon}^-}\right] - \frac{k_D}{\tilde{\varepsilon}^-}\left(\frac{\tilde{\phi}^{\mathrm{w}} \tilde{\varepsilon}^+}{\gamma_+ \gamma_-} - \frac{\tilde{\phi}^{\mathrm{w}}}{\tilde{\varepsilon}^-}\tilde{c}^+ \tilde{c}^-\right)$$

$$\tilde{p}_7 = \frac{D^+}{\tilde{\varepsilon}^+}\left[\tilde{c}^+ \frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2} - \frac{\tilde{\phi}^{\mathrm{w}} c_0^{\mathrm{F}}}{2\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\left(1 + \frac{\tilde{c}^{\mathrm{F}}}{\tilde{c}^k}\right)\right] + \frac{k_D}{\tilde{\varepsilon}^+}\left[(\tilde{c}^+)^2 \frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2}\right.$$
$$\left. - \frac{\tilde{\phi}^{\mathrm{w}} \tilde{c}^+ c_0^{\mathrm{F}}}{\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\left(1 + \frac{\tilde{c}^{\mathrm{F}}}{\tilde{c}^k}\right)\right] + \frac{k_D}{\tilde{\varepsilon}^+}\tilde{c}^+ \tilde{c}^- \frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2}$$

$$\tilde{p}_8 = \frac{D^+}{\tilde{\varepsilon}^+}\left(\frac{\tilde{\phi}^{\mathrm{w}} \tilde{\varepsilon}^-}{\tilde{c}^k \gamma_+ \gamma_-} - \frac{\tilde{\phi}^{\mathrm{w}} \tilde{c}^+}{\tilde{\varepsilon}^+}\right) + \frac{k_D}{\tilde{\varepsilon}^+}\left[\frac{2\tilde{\phi}^{\mathrm{w}} \tilde{\varepsilon}^- \tilde{c}^+}{\tilde{c}^k \gamma_+ \gamma_-} - \frac{\tilde{\phi}^{\mathrm{w}} (\tilde{c}^+)^2}{\tilde{\varepsilon}^+}\right] + \frac{k_D}{\tilde{\varepsilon}^+}\left(\frac{\tilde{\phi}^{\mathrm{w}} \tilde{\varepsilon}^-}{\gamma_+ \gamma_-} - \frac{\tilde{\phi}^{\mathrm{w}}}{\tilde{\varepsilon}^+}\tilde{c}^+ \tilde{c}^-\right)$$

$$\tilde{p}_9 = \frac{\tilde{\phi}^{\mathrm{w}} D^+}{\tilde{c}^k \gamma_+ \gamma_-} + k_D \frac{2\tilde{\phi}^{\mathrm{w}} \tilde{c}^+}{\tilde{c}^k \gamma_+ \gamma_-} + k_D \frac{\tilde{\phi}^{\mathrm{w}}}{\gamma_+ \gamma_-}$$

$$\tilde{p}_{10} = \frac{D^-}{\tilde{\varepsilon}^-}{}_B\left[\tilde{c}^- \frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2} + \frac{\tilde{\phi}^{\mathrm{w}} c_0^{\mathrm{F}}}{2\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\left(1 - \frac{\tilde{c}^{\mathrm{F}}}{\tilde{c}^k}\right)\right] + \frac{k_D}{\tilde{\varepsilon}^-}\left[(\tilde{c}^-)^2 \frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2}\right.$$
$$\left. + \frac{\tilde{\phi}^{\mathrm{w}} \tilde{c}^- c_0^{\mathrm{F}}}{\phi_0^{\mathrm{w}}(1+\tilde{e}/\phi_0^{\mathrm{w}})^2}\left(1 - \frac{\tilde{c}^{\mathrm{F}}}{\tilde{c}^k}\right)\right] + \frac{k_D}{\tilde{\varepsilon}^-}\tilde{c}^+ \tilde{c}^- \frac{\phi_0^{\mathrm{s}}}{(1+\tilde{e})^2}$$

$$\tilde{p}_{11} = \frac{\tilde{\phi}^{\mathrm{w}} D^-}{\tilde{c}^k \gamma_+ \gamma_-} + k_D \frac{2\tilde{\phi}^{\mathrm{w}} \tilde{c}^-}{\tilde{c}^k \gamma_+ \gamma_-} + k_D \frac{\tilde{\phi}^{\mathrm{w}}}{\gamma_+ \gamma_-}$$

$$\tilde{p}_{12} = \frac{D^-}{\tilde{\varepsilon}^-}\left(\frac{\tilde{\phi}^{w} \tilde{\varepsilon}^+}{\tilde{c}^k \gamma_+ \gamma_-} - \frac{\tilde{\phi}^{\mathrm{w}} \tilde{c}^-}{\tilde{\varepsilon}^-}\right) + \frac{k_D}{\tilde{\varepsilon}^-}\left[\frac{2\tilde{\phi}^{\mathrm{w}} \tilde{\varepsilon}^+ \tilde{c}^-}{\tilde{c}^k \gamma_+ \gamma_-} - \frac{\tilde{\phi}^{\mathrm{w}} (\tilde{c}^-)^2}{\tilde{\varepsilon}^-}\right] + \frac{k_D}{\tilde{\varepsilon}^-}\left(\frac{\tilde{\phi}^{\mathrm{w}} \tilde{\varepsilon}^+}{\gamma_+ \gamma_-} - \frac{\tilde{\phi}^{\mathrm{w}}}{\tilde{\varepsilon}^-}\tilde{c}^+ \tilde{c}^-\right)$$

Notice that all the quantities in the above equations are non-dimensional.

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